

GE Healthcare

Discovery CT750 HD LightSpeed VCT Installation Manual

(Book 2 of 2)



ELECTRICAL CALIBRATION, INTEGRATION & TESTING



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Rev. 5.0

Book 2 of 2: Electrical Calibration, Integration & Testing

The information in this manual applies to the following GE Healthcare LightSpeed CT Scanners:

- LightSpeed VCT (with GOC6 console)
- Discovery CT750 HD

The information in this manual does NOT apply to non-fixed (mobile) installations.

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Chapter 5

Electrical Introduction

Section 1.0 Installer/FE Notices

Section 2.0 Introduction

Use the continuity and ground checks to verify that the system power connections have not shorted to ground and that the ground and neutral connections are intact.

Section 3.0 Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your Project Manager of Installation.
- ☐ Check for short ships.
- ☐ Complete paperwork and phone calls as needed.
- ☐ Review cable connections with mechanical team.

Section 4.0 Calibration Training Requirements

See requirements outlined in Book 1.

Section 5.0 Required FE Common Tools and Supplies

TOOL	COMMENT
Standard FE Tool Kit	See new tool list
Fluke 87 DVM or equivalent	must be calibrated yearly.
AC Clamp-on amp meter	must be calibrated yearly.
Dale 600 or 601	must be calibrated yearly.
QA Phantom and phantom holder	
Lockout/Tagout kit, or equivalent	
Safety Glasses	

Table 5-1 FE Tools and Supplies

5.1 Cleanliness

The system must not be located in, near, or around construction. The room should be clean and clear of construction dust and installation materials. Do not power on the system if requirement in Book 1, Chapter 1, Section 3.13



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.

Section 6.0 Requirements/Assumptions

- The procedures in this manual are performed by an appropriately trained GE engineer.
- You need the Internet (IP) addresses the first time you execute a reconfig on the system.
 - When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.

Section 7.0 FE Workflow

- 1.) Review mechanical hand-off material. Check that the Mechanical section of GE Form e4879 is completed.
- 2.) Obtain required FE common tools and supplies.
- 3.) Perform electrical power-on and ground checks.
- 4.) Gather all customer information needed for reconfiguration.
- 5.) Perform computer integration.
- 6.) Complete Table/Gantry integration.
- 7.) Complete the calibration process.
- 8.) Perform tube warm-up and fast calibration.
- 9.) Complete tomographic plane indication.
- 10.) Check table/gantry alignment.
- 11.) Run image series tests.
- 12.) Run system functional test.
- 13.) Verify that all options were installed.
- 14.) Create system state DVD.
- 15.) Complete network connections.
- 16.) Perform Patient Touch Leakage test.
- 17.) Perform the CT System Chassis Leakage test, as required by local code.
- 18.) Complete installation and verification of any customer options.
- 19.) Complete and return GE Form e-4879 Installation Data Verification for all installations.

Section 8.0 Overview for Completing Installation

8.1 System-Level Tasks

Complete the tasks listed and check the appropriate box on the GE e-4879 form (explained in [Section 8.0 on page 215](#)).

8.1.1 General

- ☐ HVAC system is operational and environmental data reported on the GE e-4879 form.
- ☐ System realignments completed, if required.
- ☐ Broadband installed and operational.
- ☐ Power and ground audit completed.

8.1.2 Optional and Regional

- ☐ Seismic mounting kit installed, if required in your area.
- ☐ Generator recalibration completed, if necessary.
- ☐ Collimator recalibration completed, if necessary.

8.2 Site Clean-Up

- ☐ All DVDs for customer options placed in the GE service cabinet.
- ☐ All system software and service tools placed in the GE service cabinet.
- ☐ System cleaned and nicks touched up with paint.
- ☐ Room is cleaned and all trash disposed of properly.
- ☐ Installation site cleaned and all trash properly disposed.

8.3 Dolly Return

- ☐ Return of dollies arranged and dolly pick-up made.

8.4 Options

Check the appropriate box on the GE e-4879 form to verify the installation and proper functionality of all customer-ordered options.

- ☐ Injector installed and operational.
- ☐ Advantage Windows Workstation installed and functional tests completed.
- ☐ Advantage 4D installed and functional tests completed.
- ☐ Filming/Camera/DASM installed and operational.
- ☐ Modem installed and functional tests completed.
- ☐ UPS installed and functional tests completed.
- ☐ Network items installed and functional tests completed.

- ☐ Customer software options installed and operational.
- ☐ Teleradiology connections completed.
- ☐ Remote monitor installed and operational.
- ☐ Bar Code Reader installed and operational.
- ☐ Cardiac monitor and stand installed and operational.

8.5 Final Activities (Paperwork)

- ☐ GE e-4879 completed; see [Section 9.0](#). (Required for installations in ALL countries.)
- ☐ FDA 2579 completed; see [Section 9.0](#). (Required ONLY for U.S. installations.)
- ☐ Any PQRs or PSRs encountered have been reported.
- ☐ All FMIs for system completed, if necessary.
- ☐ All dispatching activities (03-04-10 codes) completed.
- ☐ Customer acceptance checks completed.
- ☐ System transfer completed and appropriate GE Healthcare personnel notified.
- ☐ All outstanding customer installation issues have been addressed.

Section 9.0 GE and Regulatory Forms

Field Engineers must complete and submit the documents listed in Section 8.1 for ALL installations, regardless of the country. In addition, for installations performed within the United States, Field Engineers must ALSO complete and submit the documents listed in Section 8.2.

9.1 All Countries

9.1.1 GE e-4879 Form

The Field Engineer should:

- 1.) Locate the GE e-4879 form on the Service CD.
- 2.) Complete the form.
- 3.) E-mail the completed form to the HHS Administrator.

9.1.2 Product Locator Cards

The Field Engineer should:

- 1.) Enter the Product Locator Card information on the Product locator Website. Go to the following address to access the site: http://gib.gehealthcare.com/gib/gib_entry.jsp
- 2.) Leave one (1) Product Locator Card (or a copy) at the customer site for EACH piece of equipment installed there.

Note: CT Manufacturing completes the GE HHS Data Sheets and provides them to the HHS Administrator.

9.2 U.S. Installations Only

9.2.1 FDA 2579 Form

The Field Engineer should:

- 1.) Download the FDA 2579 form from the HHS Support Central Web site:
http://supportcentral.ge.com/products/sup_products.asp?prod_id=16442
- 2.) Complete the form.
- 3.) E-mail the completed form to the HHS Administrator.

Note: Do NOT print this form after completion. The HHS Administrator will e-mail a printable version to the FE for customer site records.

Note: **Some states require a State Registration Number to complete this form. For any questions concerning your state, contact the HHS Administrator or check the HHS Support Central Website.**

Some states may also require additional information and test information. For instructions, contact the Project Manager of Installation.

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Chapter 6

Electrical Integration and Safety Verifications

Section 1.0 Computer Integration and Configuration

1.1 Introduction and Flowchart

This section describes the reconfiguration, system state restore, options installation, and monitor adjustment procedures.

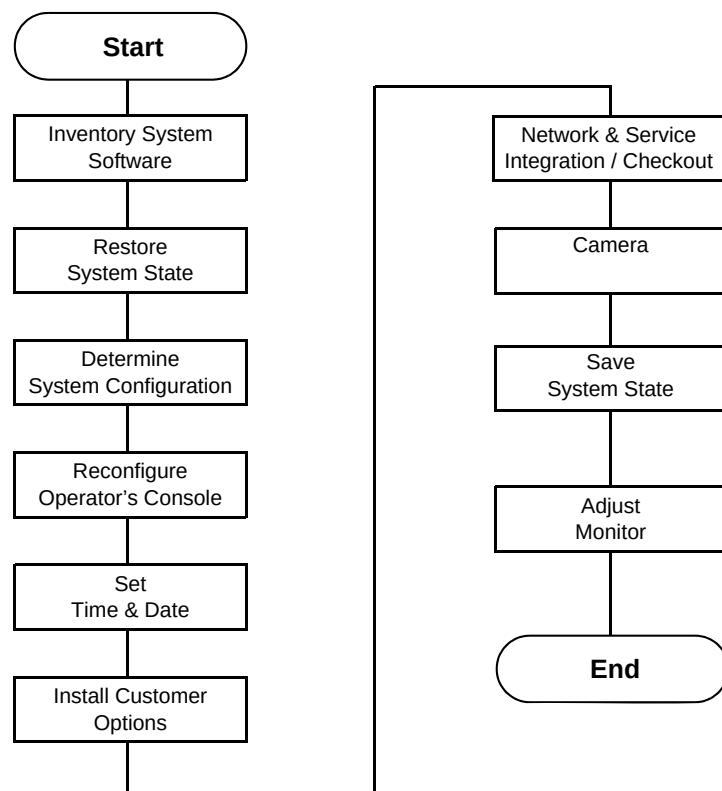


Figure 6-1 Computer Integration and Configuration Process Overview

1.2 Inventory System Software and Restore System State

Locate the box with the system software and option DVD disks, system order sheets, product locator cards, and system reconfig DVD. This tray is in the top shelf on the Lean cart labeled "Software. If not Lean packed, locate these items."

You should find the following software CD documents:

- System Operating Software CD set
- Learning and Reference Guide
- Tip Simulator
- Advanced Applications
- Service Information

Note: There may be other items in addition to those above.

1.2.1 Restore System State

1.2.1.1 Time & Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
1 (FE or mechanical supplier)			

1.2.1.2 Tools and Test Equipment

None required.

1.2.1.3 Preparation

Your system should have a system state DVD, located in the software box on the Lean cart. The system state DVD contains:

- Characterization
- Calibrations
- Gen Cal
- Other Data

If you cannot locate the shipped system state DVD and your console data is not present, you must do a **complete recalibration** of your system. If the system data is present and your Save State disk is missing, complete a Save State now.

1.2.1.4 Procedures

The installation process uses all the system state files. At this time, use the system state DVD to restore all files.

- 1.) If you are not on the Service Desktop, click the **SERVICE DESKTOP** icon.
- 2.) Click the **UTILITIES** icon.
- 3.) Select **SYSTEM STATE**.
- 4.) Insert the DVD into the DVD drive.
- 5.) Select **CHARACTERIZATION AND CAL**.

- 6.) Select **CALS**.

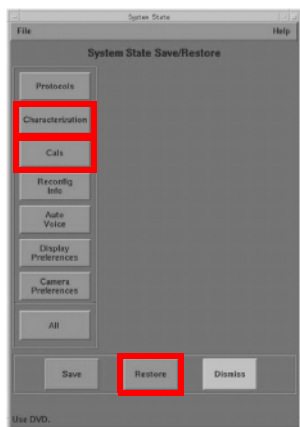


Figure 6-2 System State Restore

- 7.) Select **RESTORE** to restore the system characterization and phantom calibration files to the system.

Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.

If any error should occur during the restore process, see the Software Load Procedure manual (Load From Cold) for information regarding possible error messages and their recovery.

- 8.) Click **NO** for the Reset Scan Hardware popup message.
9.) Click **DISMISS**.

1.3 Determine System Configuration

1.3.1 Preparation

For convenient removal and use during installation, System Configuration Data Sheets are located in [Appendix D on page 301](#).

Below is a summary of key information, some of which is required from the customer, to complete system configuration. When gathering this information, refer to [Configure Site Specific Set Up on page 225](#).

System File Information:

- Hospital name (Ask the customer for ALL related fields.) _____
- Service ID _____

Patient Info:

- Next MOD# _____
- Exam #, Diagnostic # 50000 default _____
- Click **YES** to regenerate database.
- Click **NO** for Mobile System.
- HIPAA _____

Preference File Information:

- Doctor's title _____
- Date Format _____
- Time Format _____
- Language type _____
- Selected Fast Cal KV's – default - ALL unless instructed otherwise _____
- Dose Information – default - Unless instructed otherwise _____
- Dicom – default - Unless instructed otherwise _____

Hardware File Information:

- Select table type GT 2000 or GT 1700 _____
- Default for all others _____
- Network printer – default _____

Network file Information:

- Suite Name – (from FE or hospital) _____
- Host Name – (from FE or hospital) _____
- IP Address _____
- Net Mask _____
- Broadcast Address _____
- Default Gateway _____
- Advanced options – default - Unless instructed otherwise by the FE _____

1.4 Configure Site Specific Set Up

Note: The document collector box that arrived with your system contains the *Software Installation Procedures* manual, which documents the reconfiguration procedure in more detail.

1.4.1 Preparation

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the **ACCEPT** button to start the reconfiguration process, or press the **QUIT** button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged in:

`/var/adm/install.log.YYYYMMDDWWHHMMSS`

Where

YYYYMMDDWWHHMMSS is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Press the **QUIT** button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the **ACCEPT** button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

1.4.2 Procedure

- 1.) Shut down applications from the Service Desktop.
- 2.) In an **xterm** window, log in as root:
 - a.) Type: `su -` ENTER
 - b.) Type the root password; press ENTER
- 3.) Launch the Install utility:

Type: `reconfig` ENTER at the prompt.

The OC displays the Install Utility Window as shown in [Figure 6-3](#).

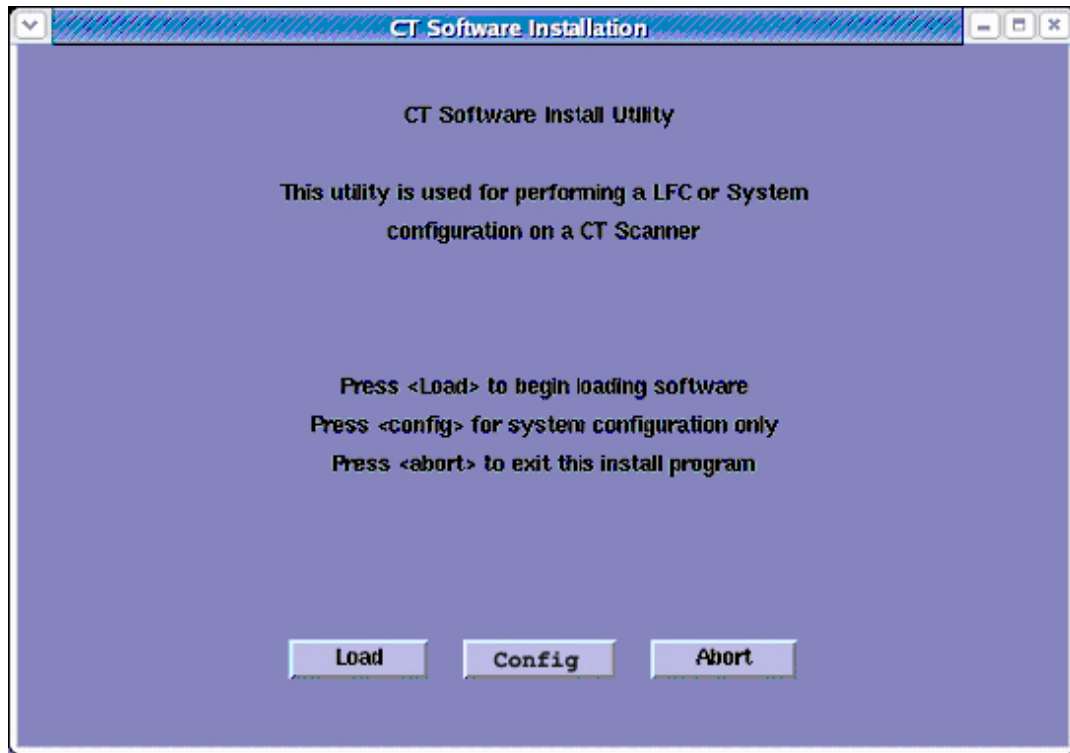


Figure 6-3 Install Utility Window

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens vary depending on the current configuration of your system.

4.) Click the **CONFIG** button.

The OC displays the System Configuration - System Settings Screen as shown in [Figure 6-4](#).

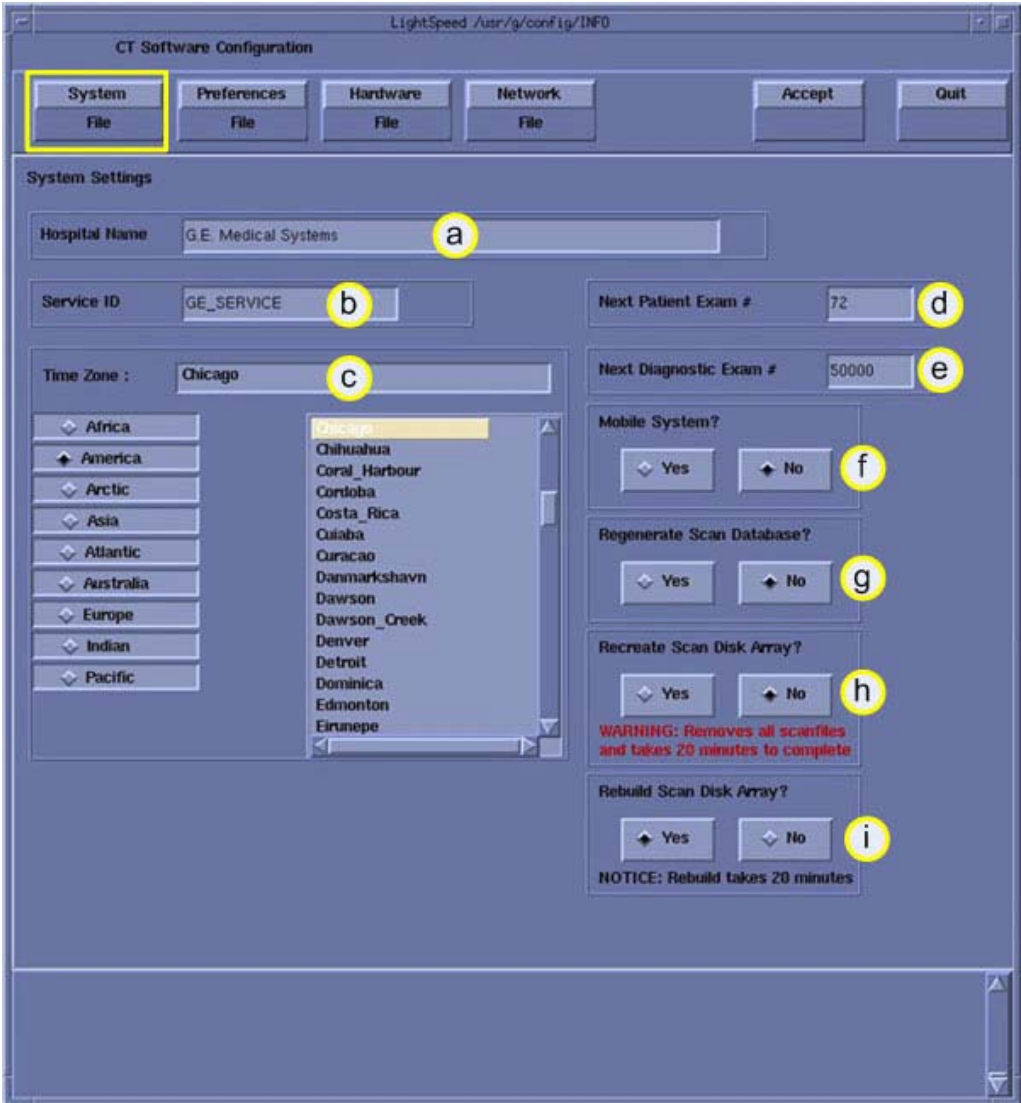


Figure 6-4 System Settings Screen

ID	Item	Description
a	Hospital Name	Configures the name that appears on images produced by this scanner. <i>Example:</i> St Marys Hospital
b	Service ID	Issued by the service organization. <i>Example:</i> 262785CT2 (no spaces)
c	Time Zone	Selects the time zone for this site
d	Next Patient Exam #	Configures the next exam number the scan user interface uses.
e	Next Diagnostic Exam #	Customer-selected; configures the next exam number the scan user interface uses.
f	Mobile System	Indicates to the software if this CT is in a mobile environment or not.

Table 6-1 System Setting Screen

ID	Item	Description
g	Regenerate Database	Determines whether the scan database is regenerated during reconfiguration.
h	Recreate Scan Disk Array	Determines whether the Scan Array is recreated during reconfiguration. Used only after HSDA Assembly replacement or multiple Hard Disk Drive failures
i	Rebuild Scan Disk Array	Used only if replacing a single hard Disk Drive in the High Speed Disk Array (HSDA) of GOP6 operator consoles. When a new hard drive is installed in HSDA and the reconfig utility is executed, a new set of buttons are displayed to allow the inclusion of the new hard Disk Drive into the array. This is not displayed during normal operation; it is displayed only after a new HDD is installed in the HSDA, and the HSDA recognizes it.

Table 6-1 System Setting Screen

5.) Configure System Settings:

- a.) Enter the Hospital Name.
- b.) Enter the Service ID.
- c.) Select the Time Zone for this site.
Use the scrollbar at the bottom of the time-zone selection list to view the entire description of a time-zone, to ensure that you are selecting the correct time-zone for your location.
If the time-zone of your location is not in the list, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time do not take effect. See the LFC manual for more information regarding time-zone setting and selection
- d.) At Next Patient Exam #, enter **1** (during installation only; this is customer-selected).
- e.) Next Diagnostic Exam #, enter **1** (during installation only; this is customer-selected).
- f.) Mobile System, select the correct answer for this installation site.
- g.) Regenerate database. Select **YES** if this is an installation with no customer data present.

Important: This destroys any Scan Data present.

- h.) Recreate Scan Disk Array: Not used during system installation.
- i.) Rebuild Scan Disk Array: Not used during system installation.

6.) Click the **PREFERENCES** button to display the Preference Settings Screen as shown in [Figure 6-5](#). for VCT Example and the [Figure 6-6](#) for HD Example.

VCT example below:

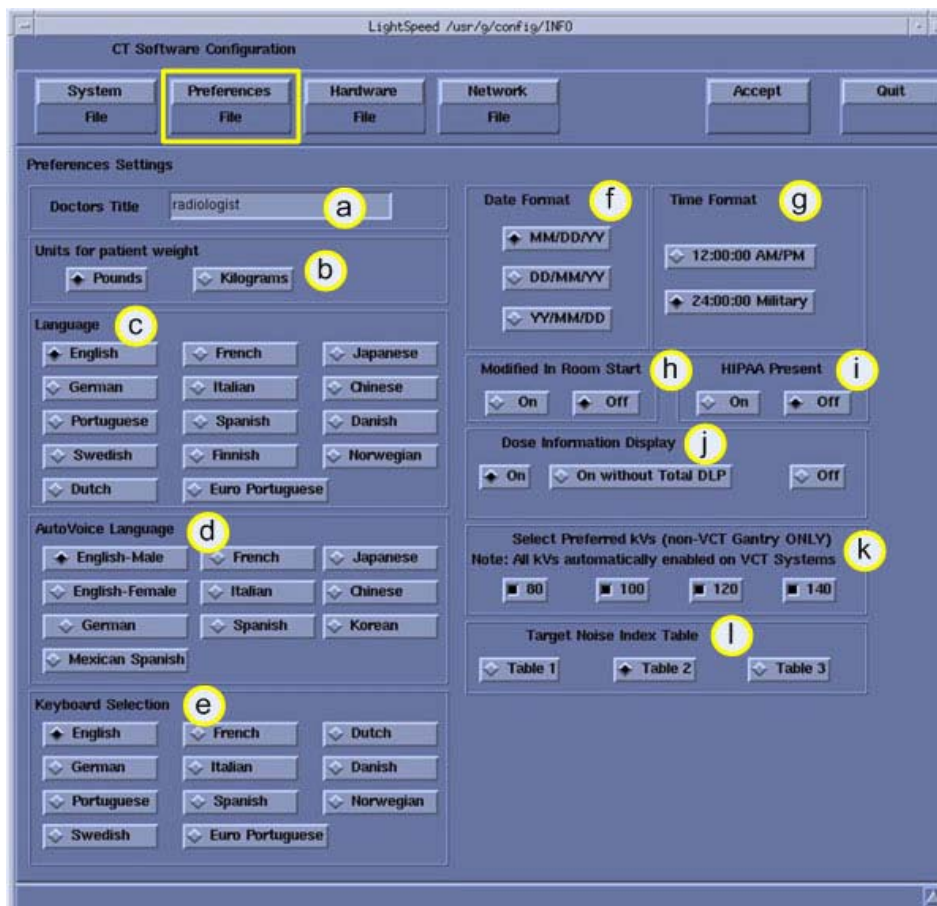


Figure 6-5 Preferences Setup Screen VCT Only)

VCT Identification		
ID	Item	Description
a	Doctor's Title	Title of the doctor (e.g. radiologist)
b	Units for Patient Weight	Identifies to the software if this site uses pounds or kilograms.
c	Language	Selects the language to display on the Application screens.
d	Autovoice Language	Configures the language heard in the scan room.
e	Keyboard Selection	Configures the language specific keyboard character set.
f	Date Format	Configures the format to display the date on the images.
g	Time Format	Configures the format to display the time on the images.
h	Modified in Room Start	Be sure OFF is selected. If this site is in Japan, select ON.
i	HIPPA Present	Be sure OFF is selected, unless told differently
j	Dose Information Display	Option for the site to use in monitoring calculated patient dose.
k	Preferred Fast Cal kV	Configures the preferred kV that the Fast Cal routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal kV field) Default is ON for VCT systems.
l	Target Noise Index Table	Select Table 2 .

Table 6-2 Preferences Settings VCT

HD example below:

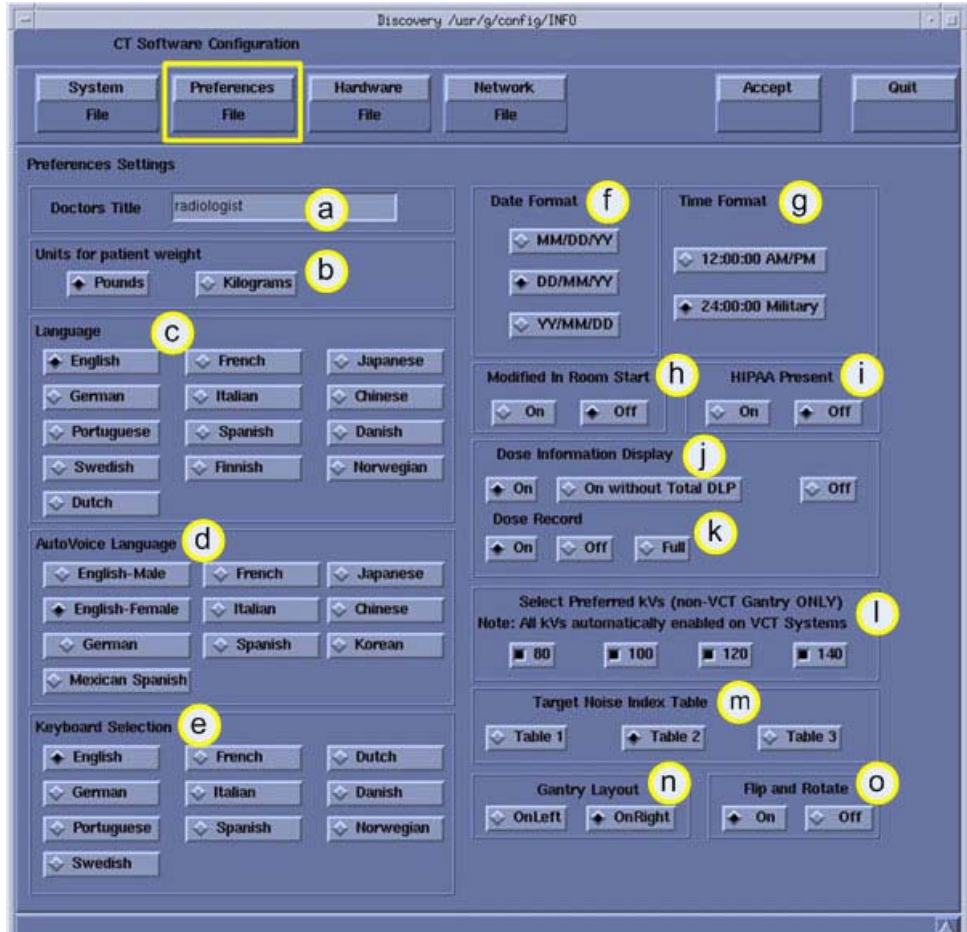


Figure 6-6 Preferences Setup Screen (HD Example)

HD Identification		
ID	Item	Description
a	Doctor's Title	Title of the doctor (e.g. radiologist)
b	Units for Patient Weight	Identifies to the software if this site uses pounds or kilograms.
c	Language	Selects the language to display on the Application screens.
d	Autovoice Language	Configures the language heard in the scan room.
e	Keyboard Selection	Configures the language specific keyboard character set.
f	Date Format	Configures the format in which the date will be displayed on the images.
g	Time Format	Configures the format in which the time will be displayed on the images.
h	Modified In Room Start	Be sure OFF is selected, unless the site is in Japan, in which case, this feature should be ON.
i	HIPAA Present	Be sure OFF is selected unless told differently.

Table 6-3 Preferences Settings HD

HD Identification		
ID	Item	Description
j	Dose Information Display	Option for the site to use in monitoring calculated patient dose. Use the default selection unless told differently. Select ON (full CTDiw Display); Select ON WITHOUT TOTAL DLP (no Dose Length Product Display); Select OFF (no CTDiw Display or Dose Report, series 999 created)
k	Dose Record	Configures support for DICOM Dose SR Record option for saving dose information with study. Default is OFF. The dose information is saved in a DICOM structured report. Select ON - Saves the dose information. Select OFF- turn off the option. Select FULL- Save the does information in a DICOM X-RAY Radiation Dose SR SOP Class
l	Preferred Fast CAI KV	Configures the preferred kV that the Fast Cal Routine will calibrate. Defaulted ON for HD systems.
m	Target Noise Index Table	Be sure Table 2 is selected.
n	Gantry Layout	Configures the preference for Patient loading. Choose correct orientation depending on site specific Gantry layout. Default is ON RIGHT.
o	Flip and Rotate	Configures the preference for allowing the Flip and Rotate feature to be turned on in the User Interface on the (Left) SCAN monitor. Default is OFF.

Table 6-3 Preferences Settings HD

7.) Configure Preferences Settings:

- a.) Enter the Doctors Title.
- b.) Select the Units for Patient Weight for this installation site
- c.) Select the Language for the customer's preference for the Applications screen.**
- d.) Select the Autovoice Language for the customer's preference.**
- e.) Select the Keyboard Selection for the language specific keyboard configuration. **

** To change this setting the Radiology Manager (or equivalent) must signoff on e4879 Installation Form.

- f.) Select the Date Format for the customer's preference.
- g.) Select the Time Format for the customer's preference.
- h.) Make sure **OFF** is selected for the Modified in Room Start, unless the site is in Japan.
- i.) Select **OFF** for the HIPAA Present, unless the customer requests differently.
- j.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose. Use the default selection unless told differently.
- k.) Select the site-preferred Dose Record. Configures support for DICOM Dose SR Record option for saving dose information with study. Default is OFF. The dose information is saved in a DICOM structured report. The DICOM standard defines a new DICOM X-RAY Radiation SR SOP class, which the other systems must support. The Dose SR feature saves an exam's dose information in this format.

* ON = Saves the dose information in a DICOM Enhanced SR SOP Class

* OFF = Turns off the option

* FULL = Saves the dose information in a DICOM X-Ray Radiation Dose SR SOP Class

Note: This preference shall not be enabled unless specifically requested by the Customer and *Evaluation of Dose SR Compatibility* functional check procedure has been executed and indicates that the other hospital systems support the Dose Report SOP classes!

- l.) Select the Preferred kVs for Fast Cal.

These kVs should include all kVs that the site uses for patient scanning. The default selections are 80, 100, 120, and 140. Use the default selection unless the customer prefers others.

- m.) Verify that Table 2 is selected for Target Noise Index

- n.) Choose the correct orientation for Gantry Layout depending on the orientation of the gantry table as viewed from the operator's console. Default is ON RIGHT.

- o.) Flip and Rotate: Configures the preference for allowing the Flip and Rotate feature to be turned on in the User interface on the (Left) SCAN Moniotr. Default is OFF. This preference allows the Customer to apply custom orientation changes based on Exam Type and reconstructions methods on the DICOM images that will be transferred to PACS and related systems.

Note: This preference shall not be enabled unless specifically requested by the Customer and *Evaluation of Image Flip and Rotate Compatibility* functional check procedure has been executed and all DICOM test images pass orientation check!

- 8.) Click the **HARDWARE** button to display the Hardware Settings Screen for VCT see [Figure 6-7](#) for CT750 HD see [Figure 6-8](#)

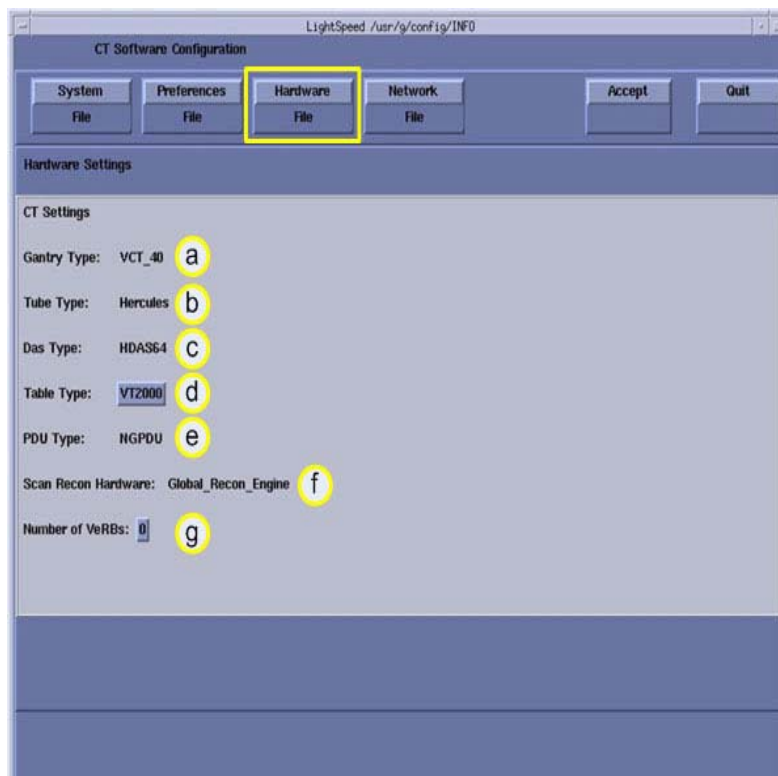


Figure 6-7 VCT Hardware Settings VCT Only

VCT		
ID	Item	Description
a	Gantry Type	Indicates the type of gantry installed with this system.
b	Tube Type	Indicates the type of X-ray tube installed in this system.
c	DAS Type	Indicates the type of DAS installed in this system
d	Table Type	Select the table type: VT 1700 (GT Long) VT 2000 (GT Short)
e	PDU Type	Indicates the type of PDU installed in this system.
f	Scan Recon Hardware	Indicates the number of VerBs installed.
g	Number of VerBs	Indicates the number of VerBs installed in this system.

Table 6-4 Hardware Settings

HD Example Below.

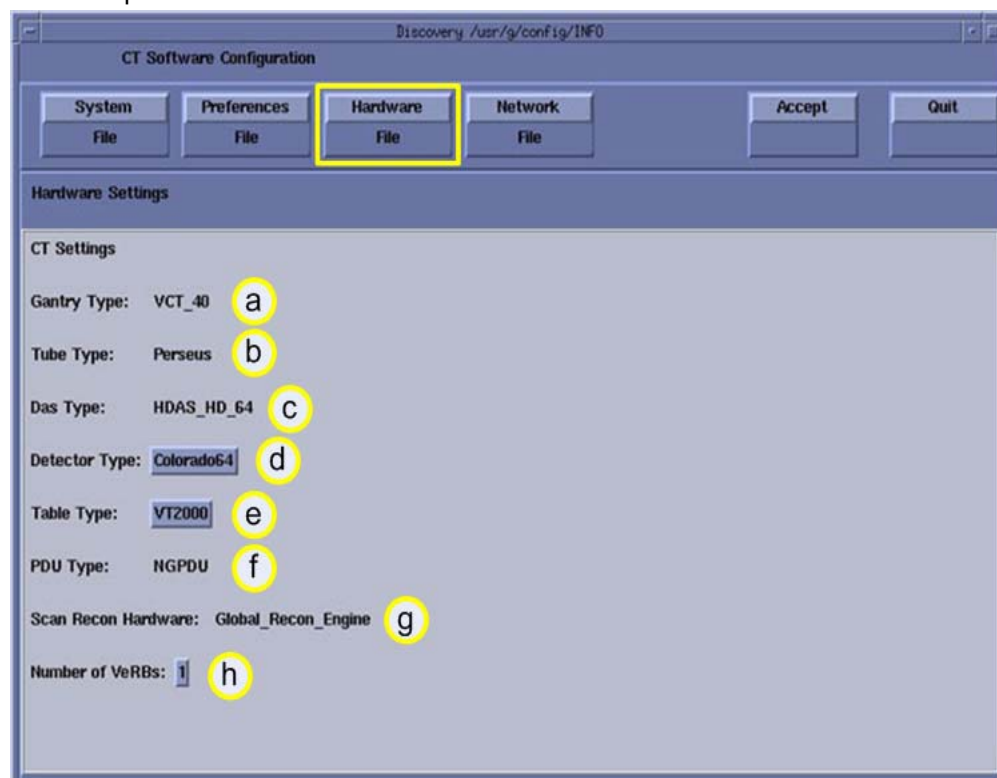


Figure 6-8 Hardware Settings Screen HD Only

CT750 HD		
ID	Item	Description
a	Gantry Type	Indicates the type of gantry installed with this system.
b	Tube Type	Indicates the type of X-ray tube installed in this system.
c	DAS Type	Indicates the type of DAS installed in this system

Table 6-5 Hardware Settings

CT750 HD		
ID	Item	Description
d	Detector Type	Select the detector type: Colorado64 for HD
e	Table Type	Select the table type: VT 1700 (GT Long) VT 2000 (GT Short)
f	PDU Type	Indicates the type of PDU installed in this system.
g	Scan Recon Hardware	Indicates console type.
h	Number of VerBs	Indicates the number of VerBs installed in this system.

Table 6-5 Hardware Settings

- 9.) Configure Hardware Settings
 - a.) Review the information for Gantry Type, Tube Type and DAS Type for this system.
 - b.) Select the Table Type installed with this system.
Determine the Table Type using the product locator card shipped with the order information.
 - c.) Review the PDU Type and Number of VerBs for this system.
- 10.) Click the **NETWORK** button to display the Network Settings Screen, as shown in [Figure 6-9](#).

Comment:

This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

See [Chapter 8](#) for more information and complete details of setting the Hospital/System Network Configuration.

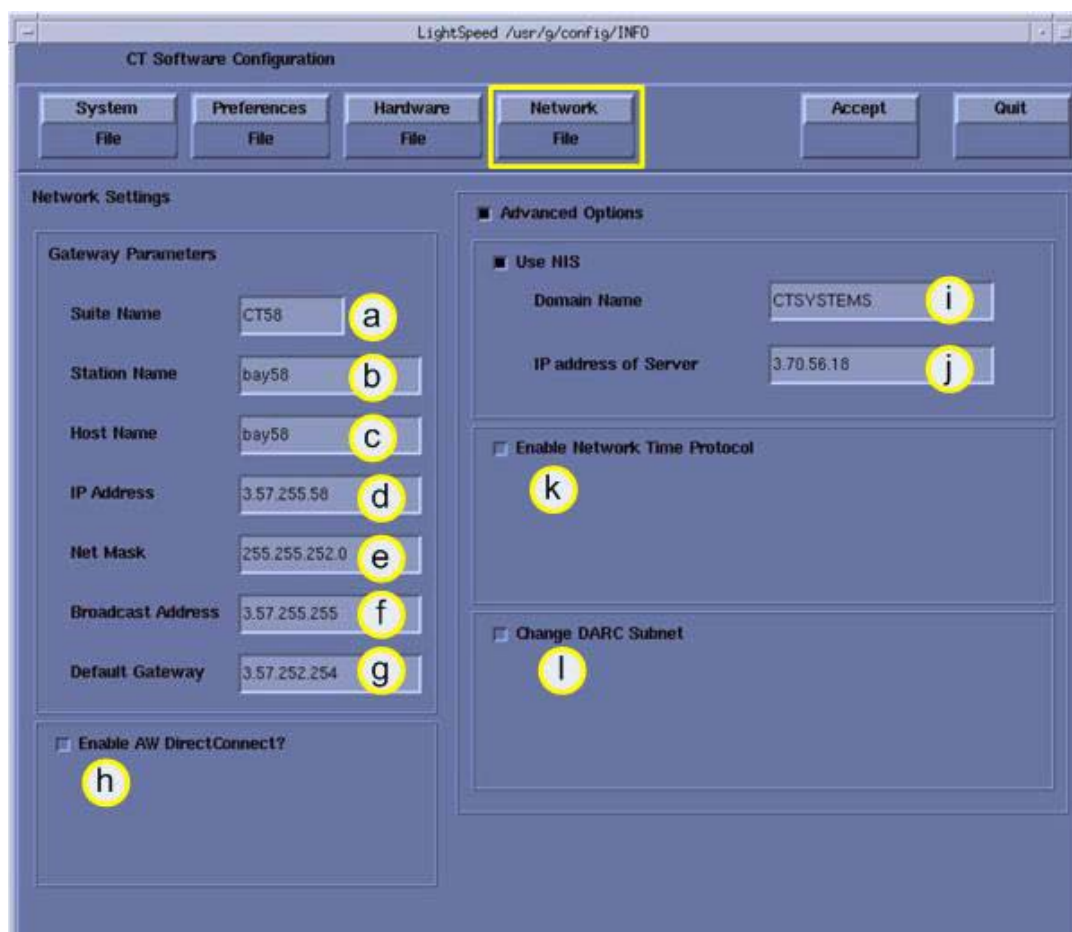


Figure 6-9 Network Settings Screen:

ID	Item	Description
a	Suite Name	The name this site is using on the system to identify the CT suite.
b	Station Name	
c	Host Name	Identifies the network hostname and AE Title of the CT system to the hospital's network
d	IP Address	Hospital's IP Address for the system.
e	Net Mask	Hospital-provided; used if the CT system is on a subnet
f	Broadcast Address	Same as the IP Address, except the last digit group is set to 1's or 0's, depending on the network configuration.
g	Default Gateway	Hospital-provided.
h	AW DirectConnect	Enable if the option is provided with the system.
i	NIS Domain Name:	Customer-provided site domain name.
j	IP Address	The IP Address for the NIS Server, if used. Hospital-provided.
k	Enable Network Time Protocol	Hospital decision.
l	Change DARC Subnet	Hospital decision.

Table 6-6 Configure Network Settings

11.) Configure Network Settings:

- a.) Enter the Suite Name.
The Suite Name must start with a letter, followed by three alphanumeric characters (a total of four characters). The name of the OC interface is <Suite Name>_OC, within the scanner's subnet. *Example: su01 or ct01* (su and ct must be lowercase)
- b.) Enter the Station Name.
 - * It cannot exceed 16 characters
 - * It can only contain **a** through **z**, and **0** through **9**.
Example: stmary or ct01
- c.) Enter the Host Name.
 - * It cannot exceed 16 characters
 - * It can only contain **a** through **z**, and **0** through **9**.
Example: stmary or ct01
- d.) Enter the IP Address.
- e.) Enter the Net Mask, if the CT system is on a subnet.
- f.) Enter the Broadcast Address
- g.) Enter the Default Gateway IP Address.
- h.) Enable the AW DirectConnect, if this option is provided with this system.
- i.) Enter the hospital-provided NIS Domain Name for the system, if NIS is utilized on-site.
- j.) Enter the hospital-provided IP Address for NIS Service for the system, if NIS is utilized on-site.
- k.) Check the Enable Network Time Protocol box, if instructed to do so by the hospital.
- l.) Check the Change DARC Subnet box, if instructed to do so by the hospital.
- 12.) Review all screens to be sure the information is correct before proceeding to the next step.
- 13.) Click the **ACCEPT** button.



Figure 6-10 Accept Button

- 14.) When the configuration changes are complete, the system displays a prompt to reboot. Click on **YES**. (See [Figure 6-11.](#))



Figure 6-11 Reboot Screen

- 15.) The system automatically logs in as ctuser after the reboot. Select **OK** on the Autostart Disabled popup message.

- 16.) Open a Shell window.

1.5 Set Time and Date

Important: You must set the time and date on the Host Computer with Application Software down.

- 1.) Open a Unix Shell and log in as root:
 - a.) Type: **su -** ENTER
 - b.) Type the root password; press ENTER
- 2.) Set the date and time for your time zone by updating the fields in the *setdate* routine.
 - a.) Type the following: **{root@hostname}# setdate**

Note: Type **q** to quit at any time; press ENTER to proceed.

To be accurate, this tool prompts you to enter the *Second*. Watch your clock or PC carefully to enter the proper value, and press ENTER at the right second to set the accurate time. Press ENTER to proceed.

- b.) Type the current Year (1980-2030).
 - c.) Type the current Month (1-12).
 - d.) Type the current Day (1-30).
 - e.) Type the current Hour (Military time, 0-23).
 - f.) Type the current Minute (0-59).
 - g.) Type the current Second (0-59).
- \Updating the time on the OC and DARC, Please Wait...
- Ping darc (172.16.0.2) 56(84) bytes of data.
- 3.) Boot system to application level.

1.6 Install Customer Options

1.6.1 Time & Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
1 (FE or mechanical supplier))			

1.6.2 Tools and Test Equipment

Laptop

1.6.3 Preparation

- Note:**
- Your system has one or more DVDs that contain customer-purchased options. Standard options are required for system operation. Install the Options DVD(s) at this time, following the instructions in this section. A more complete listing of options are located on the Service CD-ROM shipped with your system.
 - The following tables are options clarification and identification for installation purposes only.
 - Standard options are loaded before additional options. Load the standard options in the order listed in [Table 6-7](#).
 - Customer software options are located in the Option and Documentation Tray on the lean cart.

Part Number	Marketing Name	Installed Options	Comments
2371127	Smart Prep	SmartPrep	
2360576	Auto MA	AutomA	
2370921	Large Image Series	3000 Image Series	
2371407	Connect PRO	Connect Pro	Required for Exam Split
5133030	Exam Split	Exam Split	- Requires Connect Pro - Before installing, engineer needs to determine if Hard Exam Split (HES) or Virtual Exam Split (VES). - VES requires PPS (Performed Procedure Step), which is installed as a part of the Connect Pro install.
2360525	Direct 3D	Direct-3D	
5133028	DMPR	Direct-MPR	Direct-MPR requires 4 GB of Host Main memory as shown in hinv; otherwise, with 2 GB the maximum images is set to 1200.
5133032	Data Export	Data Export	
5133034	Interchange, CD/DVD	Copy Composer	

Table 6-7 Standard Options

Additional options must be loaded in the following order for proper operation.

Marketing Name	Installed Options	Comments
Neuro 3D Filter	NeuroFilter	
AutoFilter-and-Transfer	AutoFilter-and-Transfer	
HD option (64 slice)	Patient-64-Slice	
HD Hi-Power Option	HD Hi-Power Option	
Smart Score Pro	SmartScore Pro	Requires AutomA for EKG Monitor Required for Cardiac options
CardIQ Snapshot	CardIQ SnapShot	
.35 Second Rotation	Sub-0.4-Second-Scan	Requires CardIQ SnapShot
ECG Trace OC	EKG Viewer	Requires CardIQ SnapShot
Cardiac Enhancement Filter	Noise Reduction Filter	Requires CardIQ SnapShot
SnapShot Pulse	CardIQSnapShot-Cine	
DentaScan	DentaScan	
CTPerfusion3Neuro	CTPerfusion3Neuro	
Volume Viewer	VolumeViewer	Install before DMPR
CardIQPlus OR	CardIQXpressPlus OR	
Cardiac Pro on OC	CardIQ2XpressPro	Requires CardIQ SnapShot
Card EP	CardEP	
AdvVesselAnalysis	AdvVesselAnalysis	
Auto Bone	AutoBone	
CTCBase	CTColonoBase	load only one
CTCPlus	CTColonoPlus	
CTC PRO on OC	CTColonoPro	
SmartStep	SmartStep	
Volume Shuttle	AxialShuttle	

Table 6-8 Additional Options

1.6.4 Procedures

- 1.) Locate all of the option DVDs.
- 2.) Ensure that the Options DVD(s) are NOT write-protected at this time. The initial install requires that the DVD be write-enabled; subsequent installs can be done with the DVD write-protected.
- 3.) Inventory product locator cards. There should be approximately 28 cards depending on the options ordered. Use the sales order sheet that comes with the system to verify that all sold items are present.
- 4.) If you are not on the Service Desktop, click the **SERVICE DESKTOP** icon.
- 5.) Click the **CONFIGURATION** icon.
- 6.) Click **INSTALL OPTIONS**.
A blank Options window is displayed (Figure 6-12).

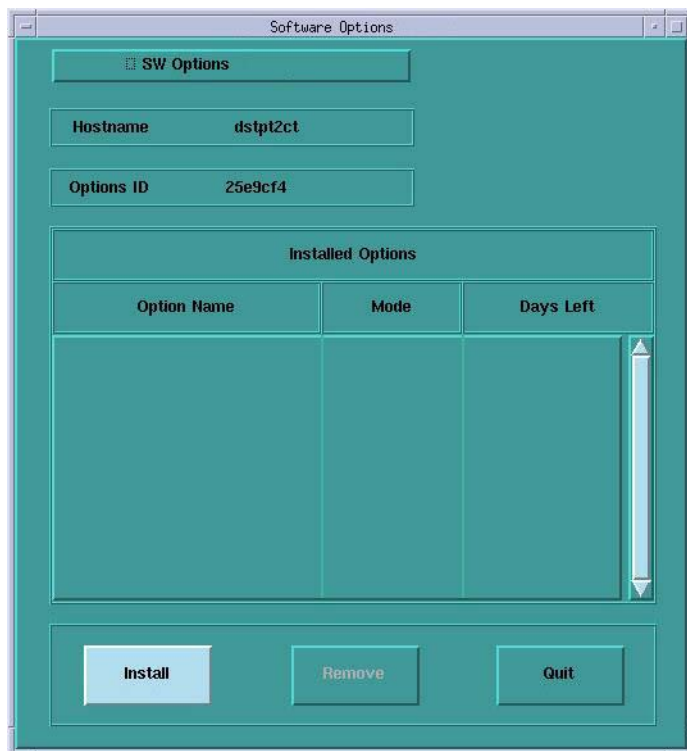


Figure 6-12 Install Options Window

7. Click **INSTALL**. A **Select Mechanism** (Figure 6-13) prompt appears. Click the button for the mechanism through which Option Keys are installed.



Figure 6-13 Select Mechanism window

- 8.) Click **PERMANENT**. A **Select Device** window (Figure 6-14) is displayed.



Figure 6-14 Options Window when First Selected

9. Click the **MEDIA** button.
- 10.) Insert the options DVD into the DVD drive and click on **OK**. (If you do not have an options DVD,

click **OK** anyway, wait for the abort pop up, then abort the process.) Available options on the DVD appear on the **Software Options** window. (See [Figure 6-15](#).)

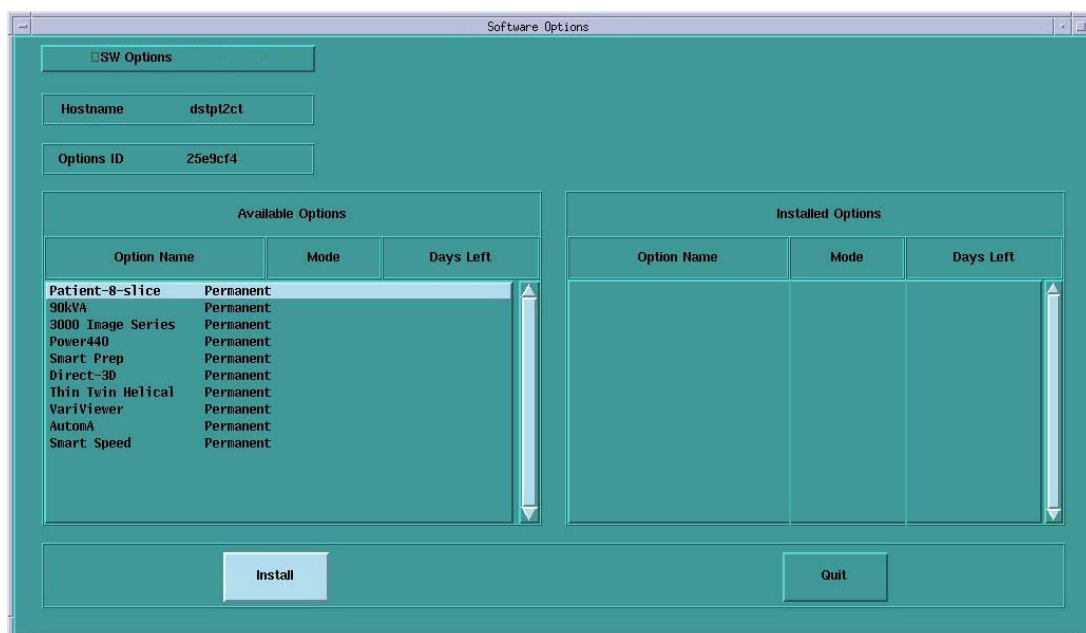


Figure 6-15 Software Options screen

- 11.) Select each option in the left-hand column in the order given in [Table 6-7](#) to install the corresponding software. The options must be loaded in the order given in [Table 6-7](#).

Note: Some options require you to complete the data field(s). Refer to the inventory sheet for the options with data requirements. Have this information available when completing this section.

- Check the FDO to see what options were ordered.
 - Compare FDO options to those on the Options DVD.
 - If different, contact your local sales representative.
- 12.) Select **INSTALL**. A box may be displayed while the options are loading. When an option is displayed in the Installed Options list on the right side, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half- minute.
 - 13.) When all options have been installed, check the permanent options on the right side of the screen against those ordered. Make changes as required and close this screen.
 - 14.) Run Verify Options to confirm that all options are loaded.
 - 15.) After the options are installed, select **QUIT** then **QUIT** again.
 - 16.) Remove the DVD and write-protect the side with options.
 - 17.) When the system prompts to reboot, click **YES**, and reboot the system to complete the installation.

1.7 Network and Service Integration and Checkout

- If additional network connections are needed for this installation, complete as required. Confirm network operation.
- If additional service integration is required to complete this installation, complete as required.
- Verify that the system information on the service home page is correct and that system service information is present on the service desktop.

1.8 Camera

1.8.1 Time & Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
1 (Field Engineer)			

1.8.2 Tools and Test Equipment

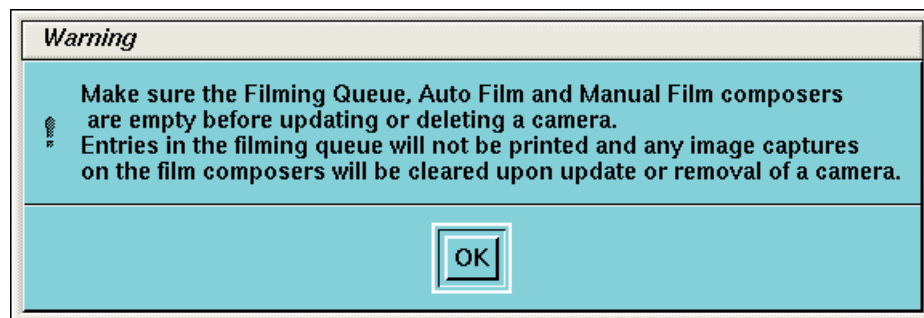
- Data collected from data sheets (See “Camera Application Configuration” on page 303.)
- Software Load Procedures manual
- System Service manual.

1.8.3 Preparation

If a DASM is required, notify the PMU that the DASM is not supported on systems with GOC6.
For details on camera configuration, refer to the Software Load Procedures manual.
For details on troubleshooting the camera, refer to the System Service manual.

1.8.4 Procedures

- 1.) Click on the **SERVICE DESKTOP** icon.
- 2.) Select **CONFIGURATION** icon.
- 3.) Select **INSTALL CAMERA**.
- 4.) Read WARNING message, and click OK.



- 5.) From the remote printer list select a camera, and select ADD for new install.
 - a.) ADD
 - b.) UPDATE
 - c.) DELETE
- 6.) Select **DICOM** or **POSTSCRIPT***.

* Follow the manufacturers suggested setup instructions..
- 7.) Follow the procedures on the screen.

Note: Camera and film information is required. Review this information with the customer. Data sheets are available in Service Information CD under Alignment, Setup and Calibrations.

- 8.) Return to the Home Page
- 9.) Click the **SERVICE DESKTOP** icon.
- 10.) Click **SHUTDOWN**
- 11.) Click **REBOOT**.

- 12.) Restart the system.

ENABLE THE CONSOLE LAPTOP PORT

- 13.) On the console in a unix shell, change user to root and enable the console laptop port:
 - a.) Type: **su -** ENTER
 - b.) Type the root password; press ENTER
 - c.) Type: **enableFEport** ENTER

1.9 Save System State and Start Up Applications

- 1.) Insert a new Save State DVD into the SCSI Tower DVD RAM drive.
- 2.) Click the **SERVICE DESKTOP** icon.
- 3.) If reloading software, click **UTILITIES**.
- 4.) Select **SYSTEM STATE**.



Figure 6-16 System State Save

- 5.) Click **ALL** to select all the cals, characterizations, etc.
- 6.) Click **SAVE**.

- 7.) If the following message appears, insert a DVD into the DVD drive and click **YES**.

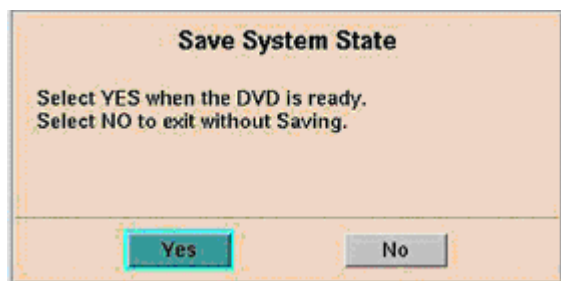


Figure 6-17 Save System State Prompt

- 8.) When completed, click **DISMISS**.
9.) Label and date the disk including the suite name.
10.) Close the Service Desktop window at the upper left corner of the screen.

Section 2.0 Table Gantry Integration

2.1 Introduction

Use these procedures to functionally check every part of the table/gantry subsystem.

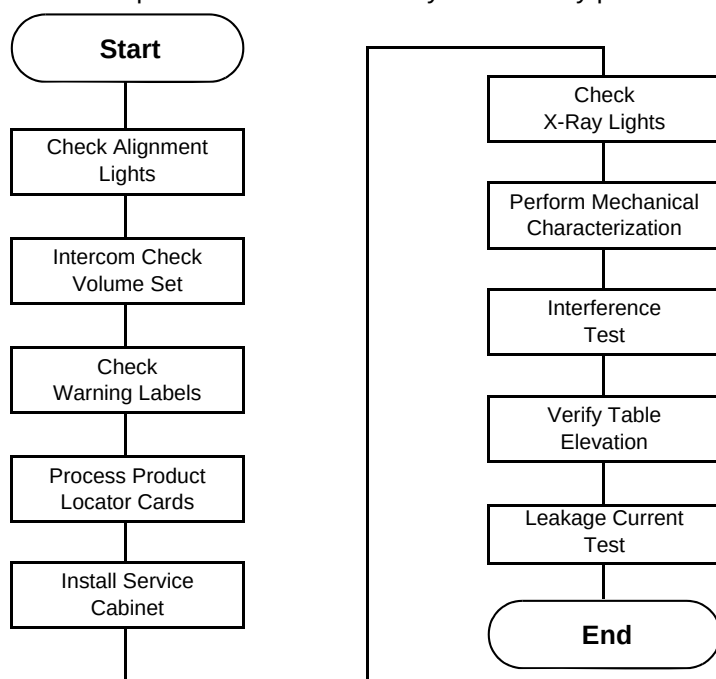


Figure 6-18 Table Gantry Integration Process Overview

Required Tool

- Multimeter

2.2 Check Alignment Lights

2.2.1 Time and Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
1 (FE or mechanical supplier)		labor on-site	

2.2.2 Tools and Test Equipment

2.2.3 Procedures



CAUTION Verify all personnel have cleared the system. The gantry rotates during this check.



- 1.) Adjust the scan room lights to normal customer operating levels.
- 2.) Turn ON the AXIAL DRIVE ENABLE and HVDC ENABLE switches (located on the gantry service switch panel).
- 3.) Turn on the alignment light switch on the gantry service panel. The gantry rotates and the alignment lights turn ON.



CAUTION **LASER EYE INJURY!**

NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE DAMAGE.

- 4.) Place a sheet of plain white paper over the output port of each light.
- 5.) Verify that the two laser lines coincide and appear as a single line.

Note: GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial lasers alignment to the collimator.

- Note:
- 6.) Ensure that cradle is level.
 - 7.) Raise the table to its highest elevation.
 - 8.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
 - 9.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals 240.0 mm $\pm$ 1.0 mm.
 - 10.) Place the rule on the left edge of the cradle, and measure again.
 - 11.) Leave the cradle in its current position, and lower the table to the minimum elevation.
 - 12.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to 240.0 mm $\pm$ 1.0 mm..
 - 13.) Press the alignment light button on the gantry control panel to turn the lights OFF.

2.2.4 Alignment Light Characterization

- 1.) Start the Mechanical Characterization tool from the Calibration tab on the Common Service Desktop.
- 2.) Select the **CHARACTERIZE ALIGNMENT LIGHTS** button from the interface.
- 3.) Follow the on-screen instructions.

2.3 Autovoice/Intercom Checks

2.3.1 Time and Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
2 people required		labor on-site	

2.3.2 Tools and Test Equipment

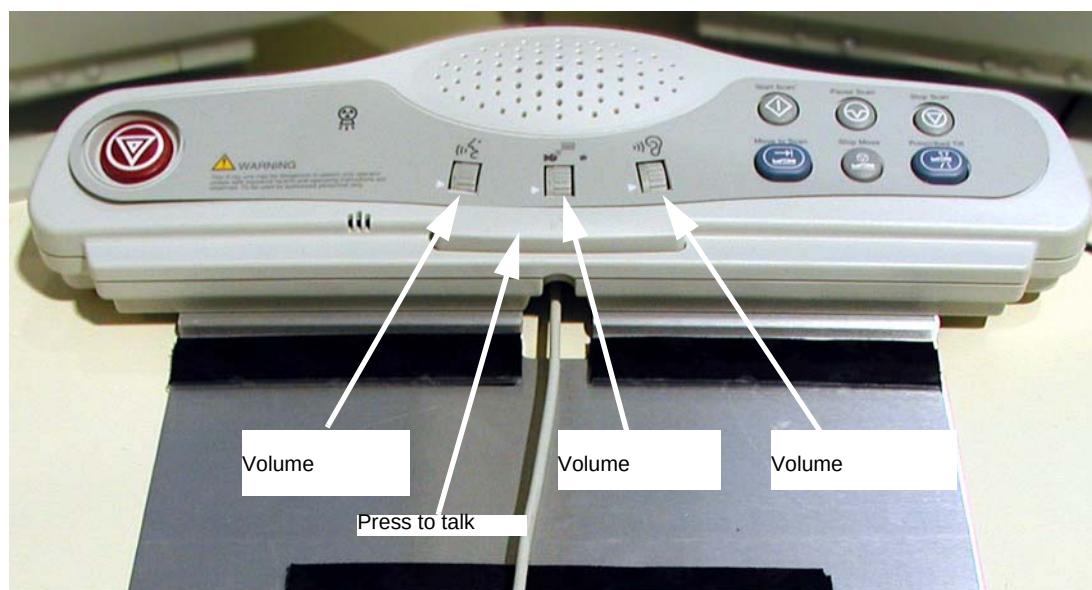


Figure 6-19 SCIM Volume Controls

2.3.3 Procedures

PATIENT SPEAKER

To adjust the volume of the patient speaker in the table:

- 1.) Adjust the left-most volume thumb wheel on the SCIM while speaking into the console microphone. (Press the bar on the SCIM to talk; release the bar to listen.)
- 2.) The assistant should be able to clearly hear the operator.

OPERATOR CONSOLE SPEAKER

To adjust the console speaker volume:

- 1.) Have an assistant speak into the gantry microphone.
- 2.) Adjust the SCIM console volume knob until you can clearly hear the assistant.

AUTOVOICE VOLUME

- 1.) On the Scan Desktop, select **PROTOCOL MANAGEMENT**.
- 2.) Select **AUTO VOICE RECORD**.

- 3.) Click the **3.4** button, to the right of "FF2. Inspiration".
- 4.) Click the **PLAY** button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5 as necessary to achieve satisfactory volume.
- 7.) Select **DONE**, then select **QUIT**.

Note: If a satisfactory volume cannot be achieved, refer to the system service manual and review the intercom module setup procedure.

2.4 CT System X-Ray ON Indicators, Cautions & Warning Labels

2.4.1 Check And Install System Warning Labels

All labels are installed in English and present on PDU, Console, Table, Gantry and Accessories. Add the labels listed below (Table 6-9 and Section 2.5) for the appropriate language for the country in which this system is installed. Additionally, apply any other warning labels if present, on equipment where appropriate.

Important: Do not cover English labels already on the system.

Subsystem	Component	Label(s)
Console	SCIM	Language overlay label X-ray warning label
	Keyboard	Keyboard warning label Function key overlay label Back warning label
Gantry	Scan Window	Laser warning label
	Laser Window	Laser warning label
	Front Cover	Laser warning label Information labels
	System GIB	System Global Installation Base (rating) label
Table (1700)	Front Side Cover	Pinch Hazard warning label - each side of cover
	Rear Side Cover	Pinch Hazard warning label - each side of cover
	Back Cradle Pan	Pinch Hazard warning label - each side of cover
Table (2000)	Front Side Cover	Pinch hazard warning label - each side of cover
	Back Cradle Pan	Pinch hazard warning label - each side of cover
NGPDU	Front Cover	Emergency OFF label Gantry Enable label Power ON label
Accessories	Table Foot Extender	Warning label
	Coronal Head Holder	Warning label
	Sagittal Head Holder	Warning label
	Accessory Tray	Warning label
	IV Pole	Caution label

Table 6-9 System Warning Labels

2.4.2 Documentation - Verification

When finished update GE Form e4879 and the installation completion form that all appropriate language labels were installed and present.

2.5 Check Warning Labels

The labels on the system and the system manuals must comply with the country law, as listed in Direction 5221102-1EN (found in the keyboard collector kit shipped with the system) regardless of the user interface (UI) language that is chosen. Compliance to the law must be completed prior to releasing the system to the customer.

Important: Do not cover English labels already on the system.

2.5.1 On SCIM

- 1.) Make sure the X-Ray warning label appears in the correct location on the SCIM.
- 2.) Record this information on GE Form e4879 located on the Service CD.

2.5.2 On Gantry

- 1.) Check that all laser warning labels are present on the gantry near the laser opening.
- 2.) There should also be warning labels on the lower right side of the gantry front cover.
- 3.) Record this information on GE Form e4879 located on the Service CD.

2.5.3 On Laser

- 1.) Make sure all laser warning labels appear in the correct location on the outside of the gantry.
- 2.) Obtain and install replacements for any missing labels.



Figure 6-20 Laser Warnings and Precautions

2.6 Process Product Locator Cards

- 1.) Collect the product locator cards shipped with the system. There should be approximately 28 product locator cards with the average system.
- 2.) Update the online product locator web site with the required hospital information.
- 3.) Confirm that the serial numbers on the cards shipped with the system match those found on the web site for that GON number. Update the information, as required.
- 4.) Place the cards in a plastic bag, then place them in the service cabinet.

2.7 Install Service Cabinet

The service cabinet is shipped assembled.

2.7.1 Time and Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
1 (FE or mechanical supplier)		1. 5 hours labor on-site	

2.7.2 Tools and Test Equipment

•

2.7.3 Procedures

- 1.) Place the cabinet in the location shown on the site print.
- 2.) Verify and place all of the service materials shipped with the system in the service cabinet.

2.8 Check X-Ray Lights

Perform several scans following the steps below. Verify that the x-ray ON lights are ON during the scans. When done, complete the information on GE form e4869 on the Service CD

- 1.) Make sure the axial drive enable and HVDC enable switches are ON.
- 2.) If you are not on the Service Desktop, click on the Service Desktop icon.
- 3.) Select **DIAGNOSTICS**.
- 4.) Select **DIAGNOSTIC DATA COLLECTION**.
- 5.) Set the scan time to 2.00.
- 6.) Set the kV to 80.
- 7.) Set the mA to 40.
- 8.) Press **ACCEPT RX**.
- 9.) Press **START SCAN** button when flashing.
- 10.) Record the above information on GE Form e4879 located on the Service CD.

2.9 Table Height Characterization

The relationship of table height to ISO center and internal-to-external landmarks must be characterized for proper interference matrix functionality.

Important: Do NOT perform tilt characterization.

- 1.) Select the **CHARACTERIZE TABLE HEIGHT** button from the interface.
- 2.) Follow the on-screen instructions.

Note: If the table height is less than 21 mm or greater than 25 mm, relative to ISO, you must adjust the table height using the table leveling pad and adjusters. Raise or lower all adjusters equally to achieve desired results.

2.10 Interference Test

PREREQUISITES

- Be sure that the System State was restored from DVD per [Section 1.2.1 on page 220](#).
- Reset the hardware to download the new characterization values before performing the table/gantry interference tests in this section.

CONFIGURATION

The table must:

- Be flashed with latest software.
- Have elevation, cradle, and IMS characterized.
- Be mechanically aligned to gantry.
- Have table/gantry characterization completed.
- Have the table extender installed.

TEST OUTLINE

The following tests verify the proper tilt and table interference matrix on the gantry.

- [Gantry Tilt Verification Test – Section 2.11](#)
- [Position Tilt, Move Table to Interference Limit – Section 2.11.1](#)
- [Position Table, Move Tilt to Interference Limit – Section 2.11.2](#)
- [Tilt Limits When Table Below Scan Plane Lower Limit – Section 2.11.3](#)

REQUIREMENTS

The following requirements are tested in this series of tests:

- 1.) No motion causes the table to hit the gantry (or gantry to hit the table)
- 2.) Requirement #1 includes the use of the table extender.
- 3.) No tilt motion causes the gantry tilting frame to touch the stationary base covers for any tilt angle.

LIMITATIONS

These requirements are only met when the table is NOT in service mode.

INTERPRETING TEST RESULTS

If test results indicate that elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize and/or adjust tilt speed. Perform elevation, cradle, and IMS first, then repeat the test. If it still fails, perform tilt.

2.11 Gantry Tilt Verification Test

Test	Steps	Expected Results
1	Move the cradle (and IMS) to home position. Push the Table Down gantry push-button to lower the table to the minimum height.	Elevation Display should read 585.0 ± 3 mm
2	Raise the table to the maximum height using the gantry controls. If the mechanical alignment of the table/gantry is not correct, as is often the case during manufacturing staging, this value may be out of range. Most of the following tests are still valid: Those that might have some variation are indicated in bold type .	Elevation Display should read 25.0 ± 3 mm. If mechanical alignment of the table/gantry is not correct, this value can be as low as 0 mm and as high as 40 mm.

Table 6-10 Table Elevation Tests

2.11.1 Position Tilt, Move Table to Interference Limit

The following tests verify the table interference limits at different tilt locations.

- Note:
- “S” means top of gantry tilts away from the table base.
 - “I” means top of gantry tilts toward the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table. Also, for all tilt angles used in this test, make sure that the Gantry Tilting frame covers do not touch the stationary base covers.

Test	Steps	Expected Results
1	Move cradle (and IMS) to home position and set internal landmark.	Cradle position on the display should read 0.0.
2	Raise table height to maximum height. Set the internal landmark. Move the IMS into gantry 400 mm, as displayed on gantry. Set the internal landmark a second time. Move cradle into gantry 1 m.	Table elevation on the display should read 25.0 ± 3 mm. (This value is the same as in the Gantry Tilt Verification Test #2 in Table 6-10.) Cradle position on display should read 1000.0 mm.
3	Tilt the gantry to I30.0. Lower the table until motion stops.	Table elevation on display should read 65 ± 3 mm. Tilt display should read I30.
4	Tilt the gantry to I23. Lower the table until motion stops.	Table elevation on the display should read 124 ± 3 mm. Tilt display should read I23.
5	Tilt the gantry to I20.0. Lower the table until motion stops.	Table elevation on the display should read 140 ± 3 mm. Tilt display should read I20.
6	Raise the table elevation to maximum height.	Table elevation on the display should read 25.0 ± 3 mm. (This value is the same as in Gantry Tilt Verification Test #2 in Table 6-10.)
7	Tilt the gantry to S30.0, then lower table until motion stops.	Table elevation on display should read 147 ± 3 mm. Tilt display should read S30.

Table 6-11 Position Tilt, Move Table to Interference Limit Tests

Test	Steps	Expected Results
8	Tilt the gantry to S23.0. Lower the table until motion stops.	Table elevation on display should read 175 ± 3mm. Tilt display should read S23.
9	Tilt the gantry to S20.0. Lower the table until motion stops.	Table elevation on display should read 185 ± 3mm. Tilt display should read S20.
10	Raise the table to 145 mm.	Table elevation on display should read 145 mm. Tilt display should read S3.0.
11	Tilt the gantry to S3.0. Verify that the table height can be adjusted from 145 mm to 25 mm. (This value is the same as in Gantry Tilt Verification Test #2 Table 6-10.)	Tilt display should read S3.0. The table lower limit should be 145 ± 3mm. Upper table limit should be 25 ± 3 mm. (This value is the same as in the Gantry Tilt Verification Test #2 Table 6-10.)
12	Set the table height to 63 mm.	Table elevation on display should read 63 mm.
13	Tilt the gantry to I30. Verify that the table height can be adjusted from 63 mm to 25 mm. (This value is the same as in Gantry Tilt Verification Test #2 Table 6-10.)	Tilt display should read I30. The table lower limit should be 63 ± 3 mm. The upper table limit should be 25 ± 3 mm. (This value is the same as in Gantry Tilt Verification Test #2 Table 6-10.)

Table 6-11 Position Tilt, Move Table to Interference Limit Tests (Continued)

2.11.2 Position Table, Move Tilt to Interference Limit

The following tests verify the tilt interference limits at different table heights.

- Note:
- “I” means top of gantry tilts toward the table base
 - “S” means top of gantry tilts away from the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

Test	Steps	Expected Results
1	Move cradle (and IMS) to home position. Set internal landmark. Set gantry tilt to zero.	Cradle position on the display should read 0.0. Gantry tilt on the display should read 0.0.
2	Raise the table to maximum height. Set the internal landmark. Move the cradle into gantry 1 m.	Table elevation on the display should read 25.0 ± 3mm. (This value is the same as Gantry Tilt Verification Test #2 in Table 6-10.) Cradle position on the display should read 1000.0 mm.
3	Lower the table until height is 115 mm. Tilt the gantry top away from the table (“S”) until it stops.	Table elevation on the display should read 115 mm. The tilt display should read S30 ± 0.5°.
4	Tilt the gantry top toward the table (“I”) until it stops.	Table elevation on display should read 115 mm. Tilt display should read I24 ± 0.5°.
5	Tilt the gantry to 0. Lower table until height is 200mm. Tilt the gantry top away from the table (“S”) until it stops.	Table elevation on display should read 200 mm. Tilt display should read S14.5 ± 0.5°.

Table 6-12 Position Table, Move Tilt to Interference Limit Tests

Test	Steps	Expected Results
6	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 200 mm. Tilt display should read $I8 \pm 0.5^\circ$.
7	Tilt the gantry to 0. Lower table until height is 210 mm. Tilt the gantry top away from the table ("S") until it stops.	Table elevation on display should read 210 mm. Tilt display should read $S10.5 \pm 0.5^\circ$.
8	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 210 mm. Tilt display should read $I6 \pm 0.5^\circ$.

Table 6-12 Position Table, Move Tilt to Interference Limit Tests

2.11.3 Tilt Limits When Table Below Scan Plane Lower Limit

The following tests verify the table and tilt interference limits when the table height is below the scan plane.

- Note:
- “I” means top of gantry tilts toward the table base
 - “S” means top of gantry tilts away from the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

Test	Steps	Expected Results
1	Set the gantry tilt to zero. Move the cradle (and IMS) to the home position. Lower the table all the way. Set the internal landmark.	Cradle position on the display should read 0.0. Gantry tilt on the display should read 0.0. Table height should read 585 ± 3 mm. (This value is the offset from 560 mm by the same amount as the upper table limit varies from 25 mm in the Gantry Tilt Verification Test #2 in Table 6-10.)
2	Tilt the gantry forward and backward. Verify the following tilt limits: S0.5 & I30.0.	Gantry tilt on the display should read S0.5 ± 0.5°. Gantry tilt on the display should read I30.0 ± 0.5°.
3	Set the gantry tilt to 0. Using the gantry push-buttons, move the cradle in toward the gantry until it stops.	Gantry tilt on the display should read 0.0. Cradle should stop at 10.0 mm ± 6 mm from the home position. (This value is not valid if the table/gantry mechanical characterization is not correct as described in the Gantry Tilt Verification Test #2 in Table 6-10. Ensure that the front of the table is at least 25 mm from the gantry.)
4	Set the gantry tilt to zero. Move the cradle to the home position.	Cradle position on the display should read 0.0 Gantry tilt on the display should read 0.0.
5	Raise the table to a height of 386 mm. Verify the following tilt limits: S11.0 & I30.0.	Table height should read 386 mm. Gantry tilt on the display should read S11.0 ± 0.5°. Gantry tilt on the display should read I30.0 ± 0.5°.
6	Set the gantry tilt to 0. Set the internal landmark. Using the gantry push-buttons, move the cradle (and IMS) in toward the gantry until it stops.	Gantry tilt on the display should read 0.0. The cradle should stop at 141 mm ± 6 mm from the home position.
7	Set the gantry tilt to zero. Move the cradle to the home position.	Cradle position on the display should read 0.0. Gantry tilt on the display should read 0.0.
8	Raise the table to a height of 242 mm. Verify the following tilt limits: S24.0 & I30.0.	Table height should read 242 mm. Gantry tilt on the display should read S24.0 ± 0.5°. Gantry tilt on the display should read I30.0 ± 0.5°.

Table 6-13 Tilt Limits When Table Below Scan Plane Lower Limit Tests

Test	Steps	Expected Results
9	Set the gantry tilt to 0. Raise the table to 210 mm. Using the gantry push-buttons, move the cradle in toward the gantry until it stops.	Gantry tilt on the display should read 0.0. The cradle should go all the way through the gantry bore to the full-extended position, (approximately 2445 mm for the GT 2000 table).
10	Set the gantry tilt to zero. Move the cradle (and IMS) to the home position. Lower the table all the way. Set the internal landmark.	Cradle position on the display should read 0.0. Gantry tilt on the display should read 0.0. Table height should read 586 ± 3 mm. (This value is the offset from 586 mm by the same amount as the upper table limit varies from 25 mm in the Gantry Tilt Verification Test #2 Table 6-10.)
11	Tilt the gantry top toward the table to a tilt of 30 degrees	Display should read I30.
12	With the table down all the way, move the cradle in until it stops.	The cradle position should be 378.5 ± 3 mm.
13	Move the table to the home position. Raise the table to a height of 386 mm. Set the internal landmark. Move the cradle in until it stops.	Cradle position should be 403 ± 5 mm.
14	Raise the table to a height of 63 mm. Move the cradle in.	The cradle should go all the way through the gantry bore to the full-extended position, (approximately 2445 mm for the GT 2000 table).

Table 6-13 Tilt Limits When Table Below Scan Plane Lower Limit Tests (Continued)

2.12 Leakage Current Test

Follow the instructions listed in [System-Level Safety Tests on page 293](#).

Chapter 7

Image Quality




CAUTION

Shock Hazard.
Voltage Present.
No service on left side while energized.



NOTICE

Potential for Data Loss and/or Equipment Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 6](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Image Quality Process Overview Flowchart

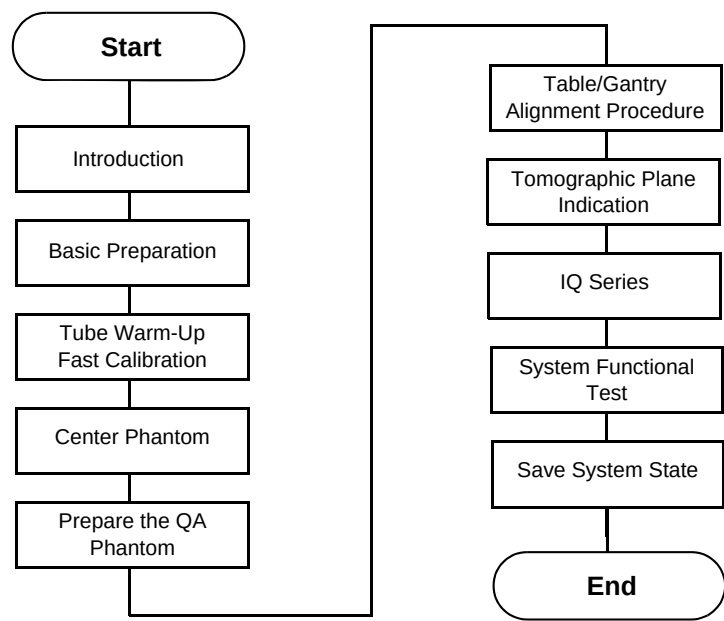


Figure 7-1 Image Quality Process Overview

Section 2.0 Reference Procedure - Scanning w/Service Protocols



NOTICE Do not perform the following procedure until instructed to do so in other sections of this chapter.

This procedure is used to locate the Manufacturing and Installation protocols on the **SERVICE** tab.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

PROCEDURE

- 1.) Select the **NEW PATIENT** icon on the left monitor.
- 2.) Enter a Patient ID (e.g., **getest**) for easy identification.
- 3.) Click on the box labeled **SERVICE** to access the necessary protocols.

Section 3.0 Tube Warm Up and Fast Cal

- 1.) Select **DAILY PREP** and **TUBE WARMUP**. Follow the steps on screen until all scans are completed.
- 2.) Select **FAST CALIBRATION** from the Daily Prep menu.

Four basic steps will be performed: Gantry Balance Check, Mylar Window Check (Warm up scans if tube is cold), Interconnectivity Map Scan List, and Fast Cal Air Scans.

Note: **Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)**

- 3.) When the calibration process completes, click on **QUIT**.

Section 4.0 Table/Gantry Alignment Procedure

4.1 Time & Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
2 (FE or mechanical supplier)		45 minutes labor on-site	

4.2 Tools and Test Equipment

- 1 mm wire

4.3 Preparation

- All table mechanical alignment procedures completed.
- The table perpendicular alignment test passed.
- Table anchors are in place and within specification.
- The table is level in all directions.

4.4 Procedure

TABLE GANTRY PREP

- 1.) Check that the table cradle is level in all directions. Correct, if necessary.
- 2.) Drive the table to its highest elevation ISO with the phantom holder removed.
- 3.) Check the scan window for proper installation.

VERIFY TABLE DRIVE CONSISTENCY

- 4.) Drive the table cradle in and out five times to seat the rollers.

CRADLE SETUP

- 5.) Turn on the alignment lights.
- 6.) Advance the end of the cradle to the black dot on cradle.
- 7.) Tape a 100 mm section of tungsten wire on the cradle that aligns with the white cradle center line.
- 8.) Using the gantry keypad, set an internal landmark, and then advance the cradle 1000 mm.
- 9.) Tape second 100 mm section of tungsten wire on the cradle that aligns with the alignment light.

SCANNING SETUP

- 10.) From the application screen Select **NEW PATIENT**.
 - a.) Fill out patient ID: **GE Test**
 - b.) Name: **Alignment**
- 11.) From the Protocol screen:
 - a.) Select **SERVICE**,
 - b.) Select **IMAGE QUALITY**,
 - c.) Select **PERPENDICULAR ALIGNMENT**.
- 12.) The red boxes should disappear from the screen. If not, reset an internal landmark.
- 13.) Select **CONFIRM**, then press the START SCAN button when lighted.

IMAGE REVIEW

- 14.) On the Service screen, select **IMAGE WORKS**
 - a.) Locate the scanned examination in the Examinations column.
 - b.) Highlight the Alignment scans

- c.) Select **VIEWER**.
 - d.) Select **FORMAT**, and select the two-in-one format horizontal display view.
- 15.) Click on image 1 and select the grid. With the grid and image displayed, visually compare image 1 to image 2.
- For close inspection, you may need to use the zoom function to see a difference.
 - Visually compare image 1 and image 2 to verify the centering wire appears in the center of the grid. As shown on the screen, the wire is 1 mm. Use the measure tool to determine the alignment difference. Move the table until both are within ± 2 mm of center.

MOVING THE TABLE

- 16.) The adjustment is likely to require a very small movement. Use a suitable tool to move the table the required distance.
- 17.) Rescan to confirm each movement trial. This can take a few trials to move the table to a position that is within the specification.
- 18.) The plastic accessory edges of the cradle are installed to allow cradle accessories to be used. These edges, although visible, should not be used to determine cradle center. Edge-to-edge difference can be greater than the alignment specification.
- 19.) Repeat above steps until both images are visually aligned on the screen.

FINALIZATION

- 20.) Use a calibrated torque wrench to tighten the anchors to 75 ± 6 N-m (55 ± 5 ft.-lb.). Confirm that the torqued anchor still meets the anchor installation specifications:
- a.) Maintain 1 full thread of adjustor showing above the lock ring or table base plate.
 - b.) Have not more than 1 in. of anchor showing above the nut. Do not cut off any access.
 - c.) Using a permanent marker, draw a line on the nut and base. Use this line to determine whether the anchor loosened over time.
- 21.) Reinstall all table components removed to access the anchors.

4.5 Tilt Alignment Tool Procedure

4.5.1 Initial Scan - Measure Alignment

- 1.) Make sure the gantry is in NORMAL (not SERVICE) mode.
- 2.) Tilt the gantry to 5 degrees in either direction, then press both tilt buttons to return the gantry to 0.0.
- 3.) Turn on alignment lights.
- 4.) Place alignment tool on the table and proceed as follows:
 - a.) Change table height so the internal coronal laser strikes the center line on the side of the

tool (Figure 7-1). This will center the tool in the FOV.

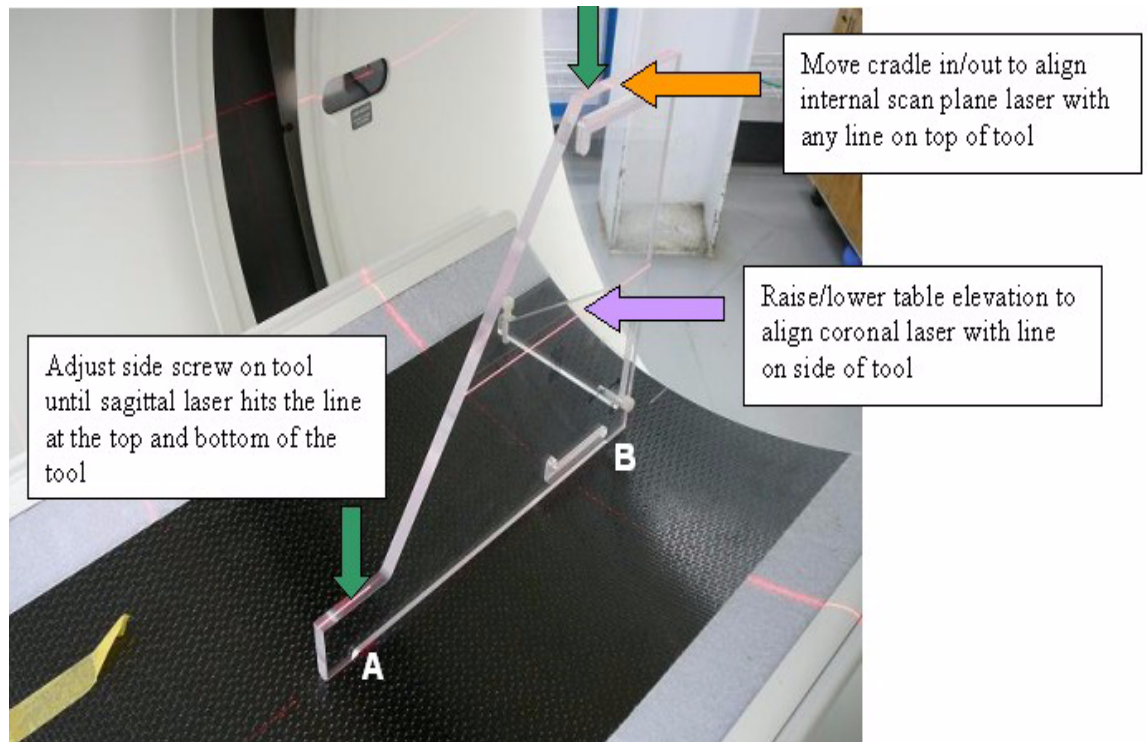


Figure 7-1 Alignment Tool on Table

- b.) Align any of the top lines with the internal laser plane by driving the cradle in/out.
- c.) Using the side adjacent screw, tilt phantom until sagittal line strikes both top and bottom lines. The reading should be 0.0.
- d.) Set an internal landmark.
- e.) Turn off lasers.
- 5.) Open a *New Patient* window at the console.
- 6.) Enter a Patient ID and Patient Name.
- 7.) Click the *Service* tab and select *Service Generic Scan*.
- 8.) Choose an Axial 1 second scan and set the following parameters:
 - a.) Thick Speed: 20.0 mm detector coverage, 0.625/32i, 1.0 rotation
 - b.) SFOV: Medium body
 - c.) Interval: 0 mm
 - d.) 120 kV
 - e.) 200 mA
 - f.) Start Location: S9.688
 - g.) End Location: I9.688
- 9.) Confirm the scan and press the [Scan] button when it lights up green.
- 10.) After the scan ends, open *Image Works* at the console and do the following:
 - a.) Select the exam and click *Viewer* on the right of the screen.
 - b.) Choose the 1-block Format to view 1 image.
 - c.) Adjust W:400 and L:-300 (using the middle mouse button)

- d.) Scroll through the images (using [Page Up] or [Page Down]) until the holes are visible (Figure 7-2).

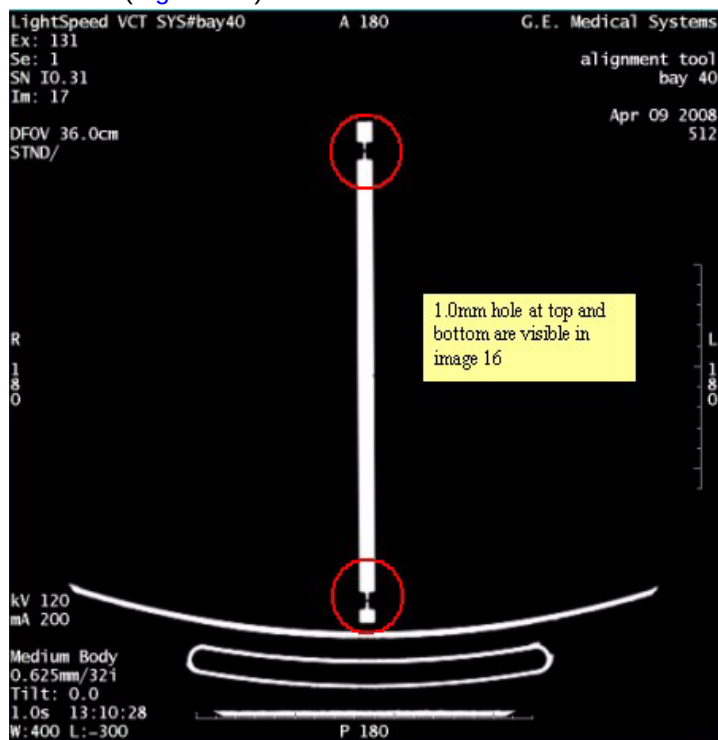


Figure 7-2 Holes Visible

(Compare Figure 7-2 with Figure 7-3, where the holes are NOT visible.)

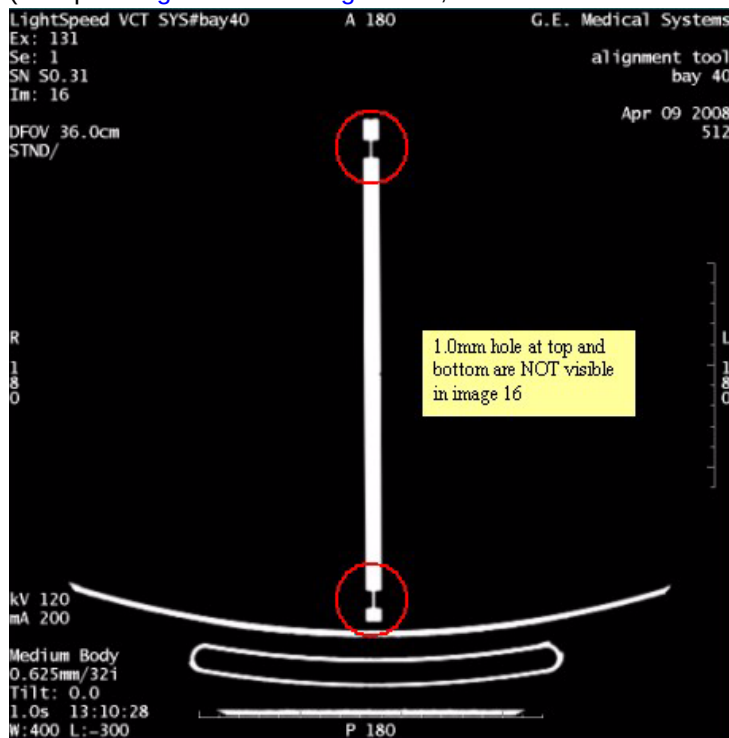


Figure 7-3 Holes Not Visible

4.5.2 Determine Number of Shims and Location

- 1.) From the images, determine the number of differences between the top and bottom hole centerlines.

Note: The holes are 1.0mm. The images are 0.625. Therefore most of the time the hole will show up in 2 images where the center line lies in between, as shown in [Figure 7-4](#). It is possible for the hole to show up in 3 images. In that case, the centerline of the hole will be in the center image and the a small portion of the hole in the outer images.

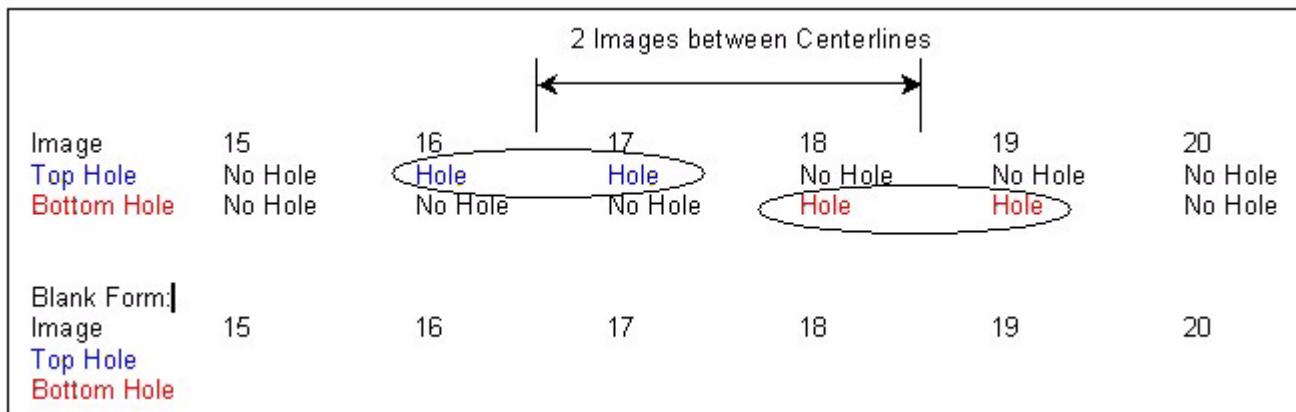


Figure 7-4 Centerline Example 1

- 2.) Using the Shim Calculator Chart below ([Figure 7-5](#)), determine the approximate number of shims. (Choose the closest number.)

Images between centerlines	Shims
0	0
1	1
2	3
3	5
4	6

Figure 7-5 Shim Calculator Chart

- 3.) Using the example below ([Figure 7-6](#)), determine on which end of the alignment tool to place

the shims.

If the top hole is to the left of the bottom hole as shown in the example below, then put the shims at point "B"						
Image	15	16	17	18	19	20
Top Hole	No Hole	Hole	Hole	No Hole	No Hole	No Hole
Bottom Hole	No Hole	No Hole	No Hole	Hole	Hole	No Hole

If the top hole is to the right of the bottom hole as shown in the example below, then put the shims at point "A"						
Image	15	16	17	18	19	20
Top Hole	No Hole	No Hole	No Hole	Hole	Hole	No Hole
Bottom Hole	No Hole	Hole	Hole	No Hole	No Hole	No Hole

Figure 7-6 Centerline Example 2

- 4.) Check phantom at center line.

4.5.3 Scan Using Shims to Verify Angle to Correct

- 1.) Using the number of shims and location determined above, repeat the scan.

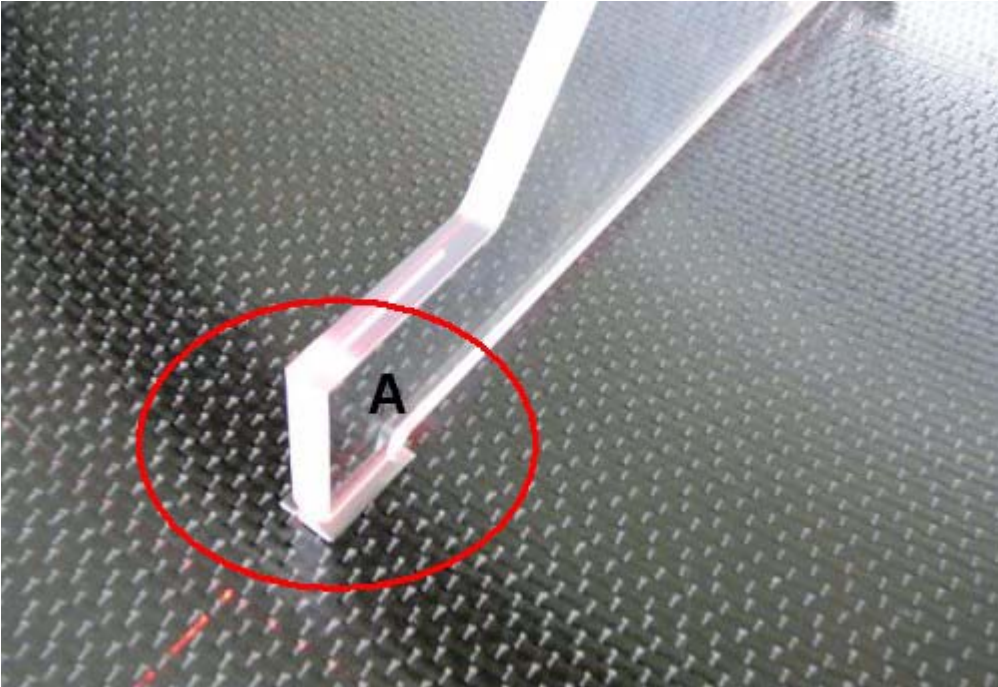


Figure 7-7 Shims Shown in Location A

- 2.) Using the previous method, verify that the top and bottom holes are off by no more than 1 image. If off by more than 1 image, adjust the shims accordingly. (See [Figure 7-8](#).)

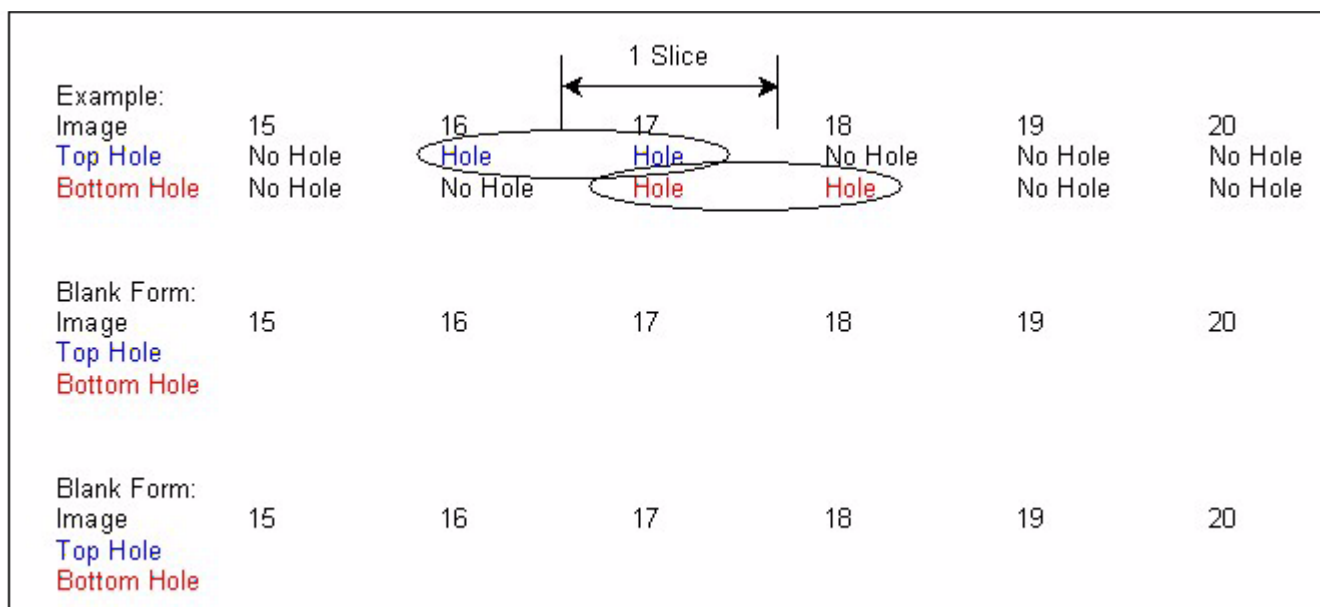


Figure 7-8 Centerline Example 3

- 3.) Ensure that alignment hasn't changed from Step 3 in the previous section, then repeat the image series.

4.5.4 Modify Table-Gantry Alignment

- Determine how much to change the gantry angle by obtaining the following information:
 - How many shims were used? _____
 - Multiply the number of shims used by 0.1 to determine angle change needed: _____
 - To which end of the alignment tool (A or B) were the shims added?: _____
- Adjust the gantry based on the chart in [Figure 7-9](#):

	Shims added at point A	Shims added at point B
Direction to turn gantry front foot levelers	Turn <u>both</u> front foot levelers <u>counter-clockwise</u> (CCW)	Turn <u>both</u> front foot levelers <u>clockwise</u> (CW)
Number of turns	½ rotation of foot = 1 shim	½ rotation of foot = 1 shim

Figure 7-9 Gantry Adjustment Chart

- 3.) Tighten the levelers.

Note: The following are technical considerations:

- Moving the front levelers clockwise will raise the front end of the gantry.
- Moving the front levelers counter-clockwise will lower the front end of the gantry.

4.5.5 Finalization

- 1.) Remove any shims from the alignment tool.
- 2.) Repeat scan as defined in [Section 4.5.1](#).
- 3.) Make additional adjustments so that the top and bottom holes can be seen in the same 2 consecutive images (0 slices between centerlines). (See Figure 7-10.)

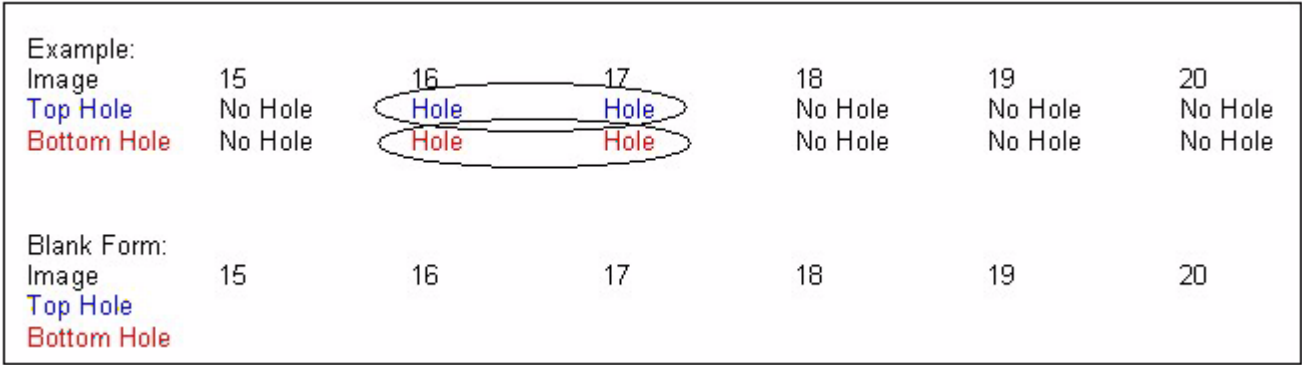


Figure 7-10 Centerline Example 4

Section 5.0 Prepare the QA Phantom

- Note: **The QA phantom is shipped water-filled.**
- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
 - 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.
 - 3.) Check for bubbles.
 - 4.) Purge and refill if necessary.

Section 6.0 Center Phantom

6.1 Required Tools

- Standard FE Tool Kit
- 9" Level

6.2 Procedure

- 1.) Locate the QA phantom and mount it and the phantom holder on the table.
- 2.) Use the adjustment knobs on the phantom holder to level the phantom front-to-back and side-to-side with a bubble level.
- 3.) Select **SCANNER UTILITIES** on the left monitor.
- 4.) Select **CENTER PHANTOM**.
- 5.) Follow the on-screen procedures.

- Note: The phantom center spec is $\pm 0.5\text{mm}$.
- 6.) Select **QUIT** when the phantom is within specification.
Level the phantom both front-to-back and side-to-side. (Use a 6" level.)

Figure 7-11

Section 7.0 Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
- 3.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
- 4.) Center the phantom in the scan plane with the **CENTER PHANTOM** found in the "Scanner Utilities" on the left monitor in the lower right corner.
- 5.) Click on **CONFIRM**.
 - a.) Adjust phantom as needed to achieve $\pm 0.5\text{mm}$ in both X and Y. Use **CONFIRM** to retry each time.
 - b.) Click **DONE** and **QUIT** when centered.
- 6.) Select **NEW PATIENT**.
- 7.) Select the service protocol, **MANUFACTURING, TOMO PLANE INDICATION**. The protocol should appear as shown in Table 7-1.

Scan Type	Start Location	End Location	No. of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV	Recon Type
Helical Full 0.8 sec	I3.000	S3.200	32	0.625 10.62 0.531:1	0.200	S0.0	Small Body	120	100	2.1	25.0	Bone

Table 7-1 Tomographic Plane Protocol

- 8.) Select **IMAGE WORKS** and select Exam, Series, Image 1.
- 9.) Click on **VIEWER**.
- 10.) Select **FORMAT** and choose single image format view.

- 11.) Locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen. See [Figure 7-12](#).



Figure 7-12 Exam, Series, Image 1

- 12.) On the GE Form e4879 Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 13.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0\text{mm}$. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0\text{mm}$ requirement.
- 14.) Repeat the Tomographic Plane Indication test with the external alignment lights.
- Use the external alignment light, and press the external landmark.
 - Verify the external light lines up along the black line on both the left and right sides of the QA phantom.
 - The scan plane indication must fall within the $S0.0 \pm 1.0\text{mm}$ specification.
- 15.) Initial below.

Section 8.0 Image Quality Test

8.1 Preparation

- All table mechanical alignment procedures are completed.
- The table perpendicular alignment test passed.
- The table tilt alignment test passed.
- Table anchors are in place and within specification.

8.2 Procedures

Important: Run ALL Image Series Tests in Auto Mode.

SETUP

- 1.) Check that the table cradle is level in all directions. Correct, if necessary.
- 2.) Check the scan window for proper installation
- 3.) From the operator's desktop, select **SCANNER UTILITY**.
- 4.) Complete **COLLIMATOR CALIBRATION** and **FAST CALIBRATION**.
- 5.) Complete section 4, **CENTER PHANTOM** procedure.
- 6.) Locate the white service CD that ships with the system, located in the tech pub tray of the lean cart.
- 7.) Load the CD into the lapto drive. Look under the Functional Checks tab located on the left hand side. Run the QA procedure.
- 8.) Record all pass/fail information on the GE e4879 form.

Section 9.0 System Functional Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

9.1 Preparation

- All table mechanical alignment procedures are completed.
- The IQ test passed.
- All options are installed and operational.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur.
- Record failure information in the comment section of the GE Form e4879.

9.1.1 What is Tested

- Scanning modes, including Cardiac.
- Installed hardware options, including:
 - UPS operation
 - cardiac monitor
 - remote monitor
 - bar code Reader
 - image transfer (if the network is operational)
 - confirm that customer network features are operational, based on options ordered and the network.
 - modem, if it is installed
 - saving an image to DVD
 - injector functional tests
 - AW functional tests
 - filming/camera functional tests
 - SmartStep

9.2 Procedure

- 1.) Locate the white service CD that ships with the system, located in the tech pub tray of the lean cart.
- 2.) Load the CD into the laptop drive. Look under the Functional Checks tab located on the left hand side. Run the System Scanning Test procedure.
- 3.) Record all pass/fail information on the GE e4879 form.

9.2.1 Confirm the following:

- 1.) The UPS goes into a backup mode when the power fails. Confirm that you can shut down the console and the table is operational before the loss of battery power.
Follow the procedure in the UPS startup/shutdown procedure.
- 2.) The remote monitor displays the same images as shown on the console image monitor.
- 3.) The bar code reader is operational and data is displayed.
 - From the **APPLICATION** screen, select **NEW PATIENT**.
 - Enter the Patient ID.
 - Using the barcode reader, scan a bar code to verify the information shown here.



Figure 7-13 Barcode Reader

- 4.) If installed, confirm the following:
- An image can be filmed, saved to DVD, and networked to another network device or AWW.
 - The Nomoto injector is operational, and all setup and calibration procedures are completed.
 - All other installed injectors must power on.
 - The SmartStep option is operational. Set up a SmartStep scan and confirm that the foot switch and the hand switch operate.

9.3 Cardiac Functional Test

9.3.1 Procedure

CARDIAC MONITOR SETUP

- 1.) Install these cables on the gantry option interface panel

CABLE	VCT	CT750 HD
NEC (power cord to wall outlet)	Yes	No
IEC (power cord to gantry)	No	Yes
Cat 5 (cable between monitor and gantry)	Yes	Yes
Lemo connector (between monitor and gantry)	Yes	Yes
Ground Connection	NA	Yes

- 2.) Turn on the monitor. Follow the monitor self test setup procedure using the document shipped with the system.
- 3.) From the **APPLICATION** screen, select **NEW PATIENT**. Fill out:
- Patient ID: **GE Test**
 - Name: **Cardiac Functional System Functional Test**
 - Select from Protocol Menu
 - * User
 - * Chest

- On the dark blue bar on the scan monitor select **Gating "On"**
- 4.) Check for presence of these items:
 - a.) Heart rate on the gantry display board
 - b.) Cardiac pulses shown on the screen
 - c.) Gating BPM displayed on the screen
 - d.) ECG trace highlighted on the screen

9.4 Connect Pro Functional Test

9.4.1 Procedure

PATIENT SCHEDULER

- 1.) On the scan desktop select **Patient scheduler**
- 2.) Select the button labeled **Update**
- 3.) The customer list should be present select **Export data**
- 4.) Send a test image to a workstation **AWW or what ever is available.**

9.5 Finalization

This section is complete when all of the above items pass.
Check all appropriate boxes on the GE Form e4879.

Section 10.0 Save System State

Use the following commands to create the System State DVD.

- 1.) Insert a DVD into the DVD drive.
- 2.) If you are not on the Service Desktop, click the **SERVICE DESKTOP** icon,.
- 3.) Click on **UTILITIES** icon.
- 4.) Select **SYSTEM STATE** to open the System State Save/Restore menu.
- 5.) Select **ALL**
- 6.) Select **SAVE**
- 7.) When the save operation completes, select **FILE** and **QUIT** from the pull-down menu.
- 8.) Remove the DVD from the drive.

Chapter 8

Customer Options Installation & Verification

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 DICOM Network

1.1 Introduction

LightSpeed systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM networks operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The CT scanner system is uses six DICOM Network Services and is provides two DICOM Services:

1.1.1 As a DICOM Service User:

- Send or **"Push"** images to another network device.
- Send or **"Push"** images to a DICOM Printer.
- Review image database on another device and retrieve or **"Pull"** selected images from that device (Query/Retrieve User).
- Send or **"Push"** images to a an image storage device and obtain confirmation that the images were archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on MOD media.

1.1.2 As a DICOM Service Provider:

- Receive **"Pushed"** images from another network device.
- Allow another network device to review the image database and to retrieve or **"Pull"** selected images (Query/Retrieve Provider).

For each DICOM Service that the CT system is a **User** (except for storing images on MOD media), you must **declare** this device on the CT system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides.

For example, you may need to declare a PACS System twice on the CT system: once as a destination to **push** images and, second, as destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the CT system is a **Provider**, you must **declare** the CT system on the network device that is using these services.

Information required to complete configuring a hospital DICOM network is provided by:

- The hospital network administrator (hostnames, IP Addresses)
- The DICOM Conformance Statement document (AE Titles, Port Numbers), provided with each DICOM compatible network device on the network.

1.2 Preparation

1.2.1 Network Physical Requirements

Before setting up the CT scanner system on the hospital network, verify the following physical items are complete:

- Scanner console, monitor, keyboard, and mouse are installed and connected.
- CT system power is ON.
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the scanner console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

1.2.2 Network Identity Information

You need to gather network identity information to do the following tasks:

- Declare the CT system on the network
- Declare the DICOM remote hosts (PACS systems, archival devices, review workstations) on the CT system.
- Declare the DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the CT system.
- Declare the DICOM on the CT System,

Ensure the following network identity information is available:

- From the Hospital Network Administrator:
 - Hostname (No more than 16 Characters).
 - Internet Protocol (IP) Address.
 - Subnet Net Mask IP Address (if applicable).
 - Broadcast Address (if applicable).
 - Network Protocol (DICOM for CT Systems)
- From the Remote Host Device DICOM Conformance Statement Document:
 - DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses).
 - DICOM Listening Port Number.
- From the HIS/RIS Interface Device DICOM Conformance Statement Document:
 - DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides).
 - DICOM Listening Port Number.
- From the Printer DICOM Conformance Statement Document:
 - DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses).
 - DICOM Listening Port Number.

1.3 Procedures

1.3.1 Enter Configuration Routine

- 1.) On the operator's console, open a shell window.
- 2.) Enter root as a superuser:
At the prompt, type: **su - ENTER**
At the password prompt: type the password; press **ENTER**

- 3.) Change directory to scripts.
Type: **cd /user/g/scripts** **ENTER** at the root prompt.
- 4.) Launch the Install Utility:
At the prompt, type: **reconfig** **ENTER**
The OC displays the Install Utility Window as shown in [Figure 8-1](#).

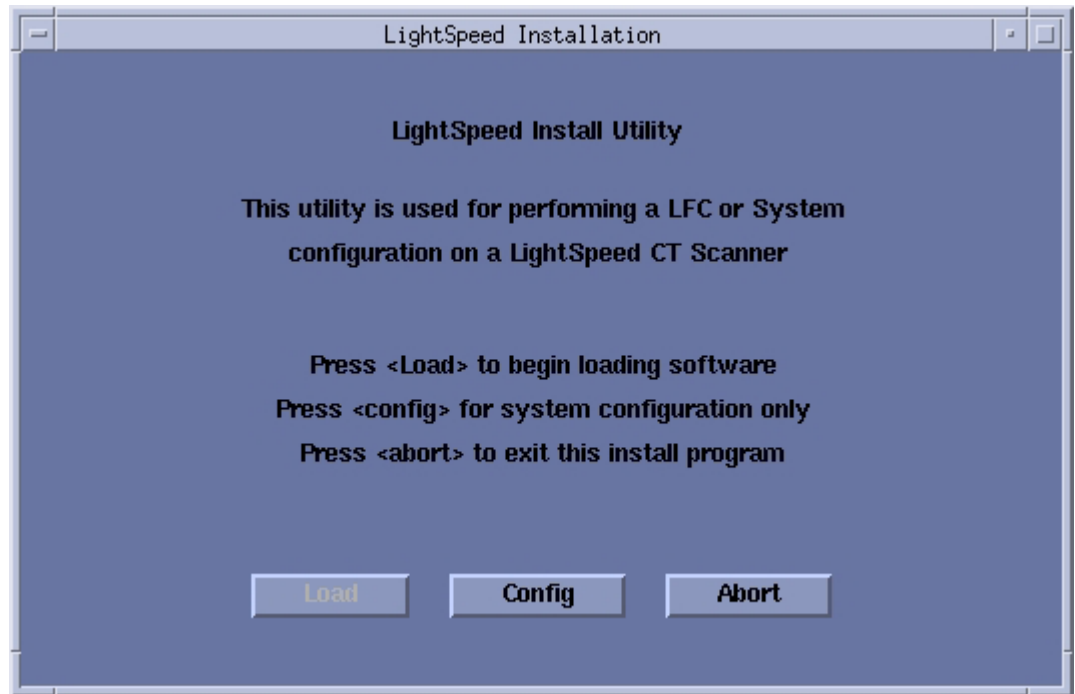


Figure 8-1 Install Utility Window

- 5.) Enter the Configuration Routine:

Click the CONFIG button.

The OC displays the System Configuration - System Settings screen, as shown in [Figure 8-2](#).

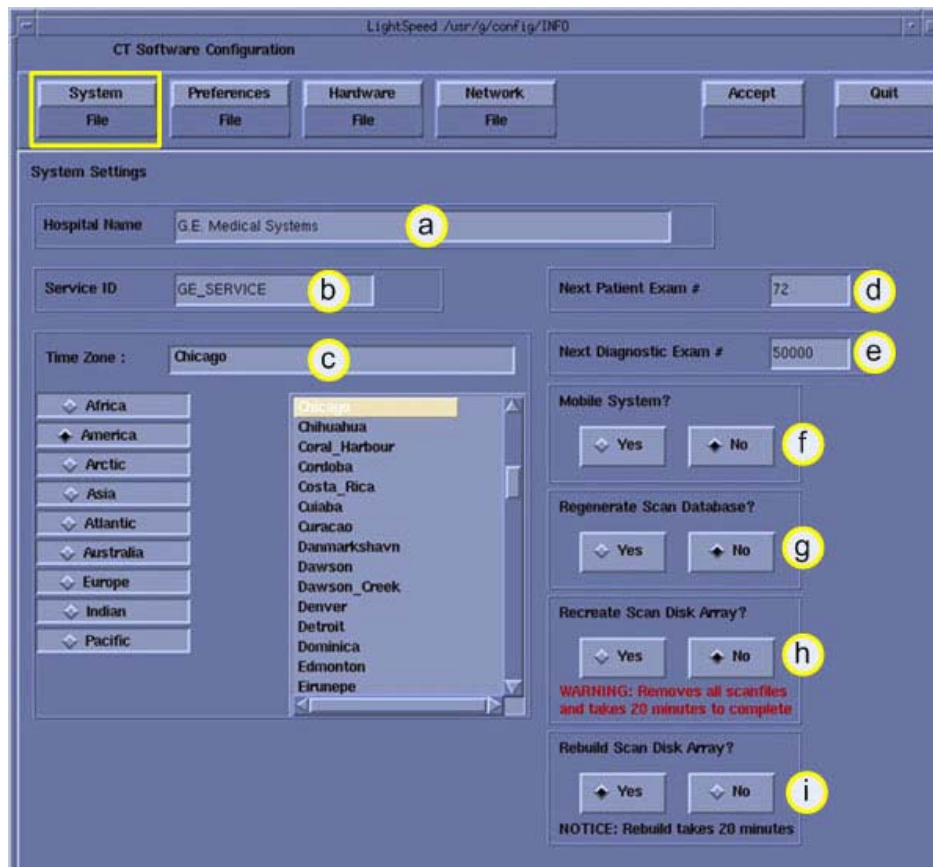


Figure 8-2 System Settings Screen

1.3.2 Configure Network Settings

This screen allows you to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 8-3](#).

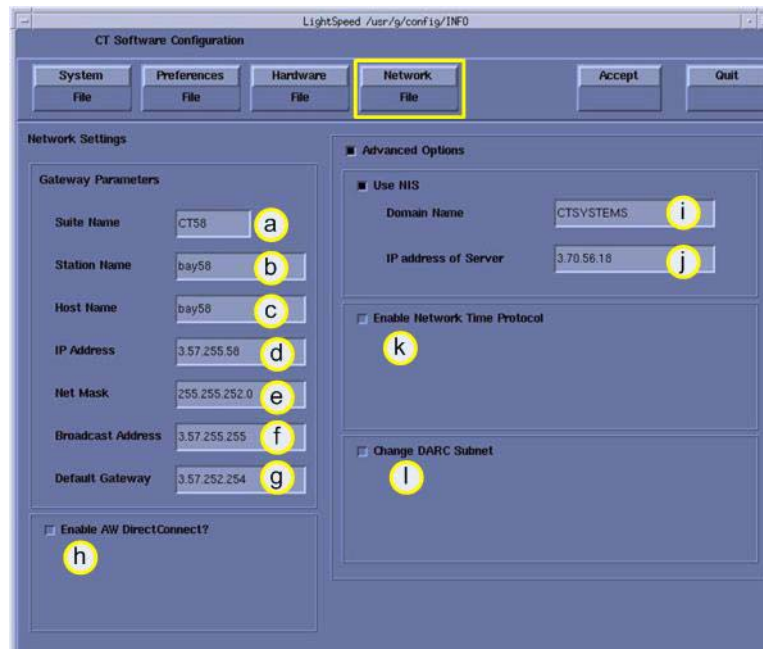


Figure 8-3 Networks Settings Screen

- 2.) Enter the Suite Name.

The Suite Name identifies this particular CT system as a part of a group of CT Systems in a suite configuration. This Suite Name appears on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface is <Suite Name>_oc and the SBC interface is <Suite Name>_sbc.

- 3.) Enter the hospital-provided Host Name.

The Host Name identifies the network hostname and AE Title of the CT system.

The Host Name:

- **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
- **MUST NOT** exceed 16 characters.
- **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _

- 4.) Enter the hospital-provided IP Address.

- 5.) Enter the hospital-provided Net Mask (if the CT system is on a subnet).

- 6.) Enter the Broadcast Address:

The Broadcast Address should be the same as the IP Address, except for the bits of the host ID portion (last digit group) set to 1's or 0's, depending on the configuration of the network. The standard default is 1's but older SunOS machines used 0's.

Example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.

If the network contains genesis-based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- 7.) Enter the hospital-provided Default Gateway IP Address (if applicable). If the site network does not use a default gateway, leave the field blank.

- 8.) Select NIS (Yellow Pages database) Advanced Option only if requested by the hospital network administrator as follows:

- a.) Select ADVANCED OPTIONS button on the Network Settings screen.

- b.) Select USE NIS? button.
- c.) Enter the hospital-provided Domain Name.
- 9.) Record all the Network parameters in the *Software Installation Procedures* Document, or on the worksheet in Appendix E, [Section 4.0 on page 303](#).

1.3.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.
The system loads the application software, OS patches, and kernel changes, and configures the system on both the OC and the SBC.
This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:
`/var/adm/install.log.YYYYMMDDWWHHMMSS.`
(Where `YYYMMDDWWHHMMSS` is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click YES.
- 3.) The system automatically logs in as ctuser after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console shell window, type: **startup** ENTER.

1.3.4 Declaring Remote Hosts on the CT System

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.

1.3.5 Declaring Advantage NET Remote Hosts on the Scanner

Use Advantage NET Protocol networks to communicate with older CT Systems (CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with LightSpeed DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host

Parameter Screen as shown in [Figure 8-4](#).

Figure 8-4 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital-provided Host name.
- 3.) Enter the hospital-provided Network Address (IP Address).
- 4.) Select ADVANTAGE NET as the Network Protocol.
The system automatically removes the highlighting from the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

1.3.6 Declaring DICOM Remote Hosts on the CT Scanner

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host

Parameter screen as shown in [Figure 8-5](#).

Figure 8-5 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.
- 6.) Enter the AE Title from the DICOM Conformance Statement provided with the device.

The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.

Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the CT System. For most devices, the AE Title is the same as the hostname (CT systems are equipped with this feature).

However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).

Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the CT system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.

- 7.) Select the correct Archive Node choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - AUTO to have the CT system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - YES if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate Archive Status = Y to the operator.
 - NO if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct Access to local host: settings. These two selections allow you to selectively block the remote host from using the LightSpeed DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - Send Images? Set to YES if the customer wants the CT system to be able to have images **pushed** to the system from the applicable remote host. Set to NO if the customer wants to block an image **push** from the applicable remote host.
 - Query/retrieve images? Set to YES if the customer wants the remote host to be able to review the image database (query) and **pull** selected images from the database. Set to NO if the customer does not want the remote host to have this ability.
- 9.) Select the correct Custom search? setting. This selection allows the CT scanner to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the CT system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select ON if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select OFF if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *Software Installation Procedures* Document.
- 11.) Select SAVE to store the parameter settings of the remote host.

1.3.7 Declaring the CT System on Remote Hosts

1.4 Declaring the Scanner on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the CT System as an Advantage NET remote host.

1.5 Declaring the Scanner on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the CT System as a DICOM remote host.

The CT System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the CT image database.
- A remote host can review the CT image database (query) and **pull** selected images (retrieve).

Use the following parameter information to configure the DICOM device/system to either **push** images to the CT scanner and/or perform a **Query/Retrieve** operation:

- Host name: Provided by the Hospital Network Administrator. Exactly the same scanner assigned hostname entered in Network Configuration Screen.

- **Application Entity Title:** Exactly the same entry as the Hostname.
- **Network Address:** Provided by the Hospital Network Administrator. Exactly the same scanner assigned IP Address entered in Network Configuration Screen.
- **Network Protocol:** DICOM 3.0.
- **Port Number:** For all DICOM service that the CT System provides, use 4006.
- **Provider Type:** This field concerns the LightSpeed DICOM **Query/Retrieve** provider capability. All CT systems are study root systems, which allow queries at the exam, series, and image level.
- **Support Worklist:** This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All CT systems support a filtered search of the image database as part of the LightSpeed DICOM **Query/Retrieve** provider capability.

Section 2.0 DICOM HIS/RIS Setup

2.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the CT system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer. In addition, the CT system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

2.2 Loading ConnectPRO Software Option on the CT System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Click on INSTALL.
- 4.) Select INSTALL OPTIONS and click START. The console displays the Software Options window as shown in [Figure 8-6](#).

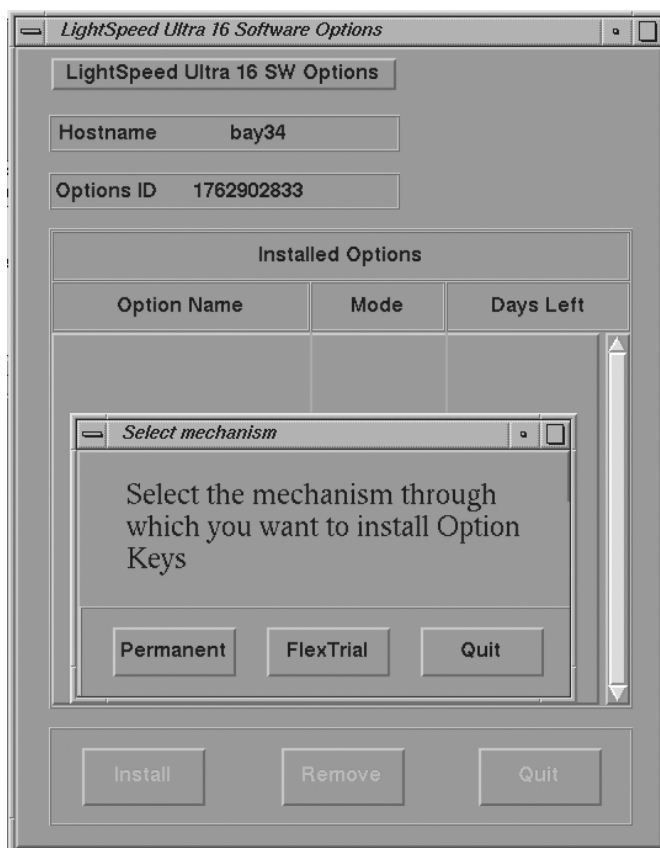


Figure 8-6 Options Window when First Selected

- 5.) Insert the ConnectPRO Options MOD into the MOD drive.
- 6.) Click on OK to continue.
- 7.) On the Software Option window, select the CONNECTPRO option in the Available Options window.
- 8.) Click on the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option, and
- the console displays the ConnectPro Setup window as shown in [Figure 8-7](#).

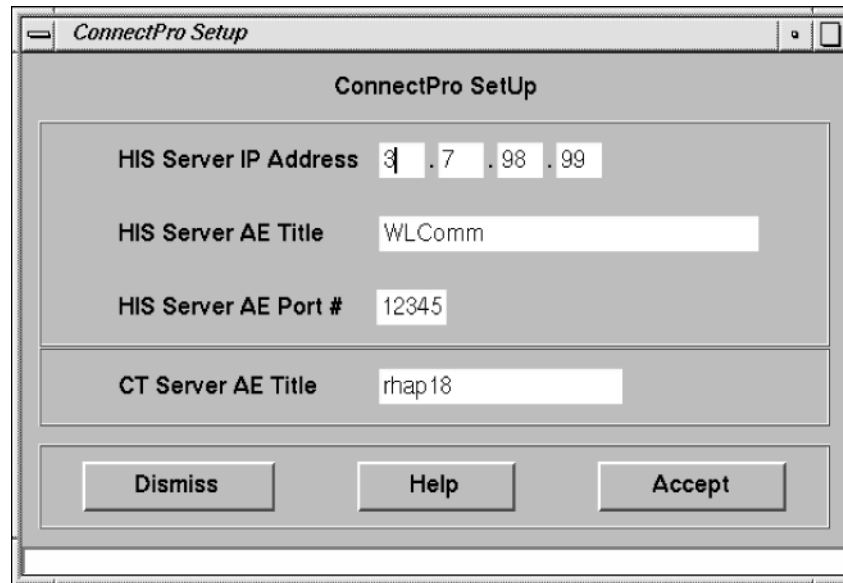


Figure 8-7 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the ConnectPRO HIS/RIS Network parameters in the *Software Installation Procedures* document.

The system will automatically load the LightSpeed CT Server AE Title (Hostname of the system).

- 13.) Click on the ACCEPT button. The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the Software Option window shown in [Figure 8-6](#).
- 14.) Click on the QUIT button and the subsequent OK button in the message window.
- 15.) Remove the MOD from the MOD drive, and write protect the side containing the ConnectPRO option.

2.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in ~ctuser/app-defaults/devices must be manually edited to remove the offending characters in the set configuration line. Invalid characters include \{ }

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.

For Trim Off - set	TRIM NO
For Trim On - set	TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for LightSpeed 7.X (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.

To force a 14x17 film size - set	filmSize 14INX17IN
----------------------------------	--------------------

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
 - To prevent sending the Smoothing Parameter set FB_Smooth FALSE
 - To prevent sending the Border Density set FB_Border FALSE
 - To prevent sending the Empty Image Density set FB_EID FALSE
 - To prevent sending the Minimum Density set FB_MinD FALSE
 - To prevent sending the Trim Parameter set FB_Trim FALSE

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the install.dasm (Association Error)
- **Solution:** The system is unable to complete the association. Check the AE Title and the Port number of the DICOM print server and correct them through the Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the prslog file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GE Healthcare Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GE Healthcare Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Section 3.0 Network Connections

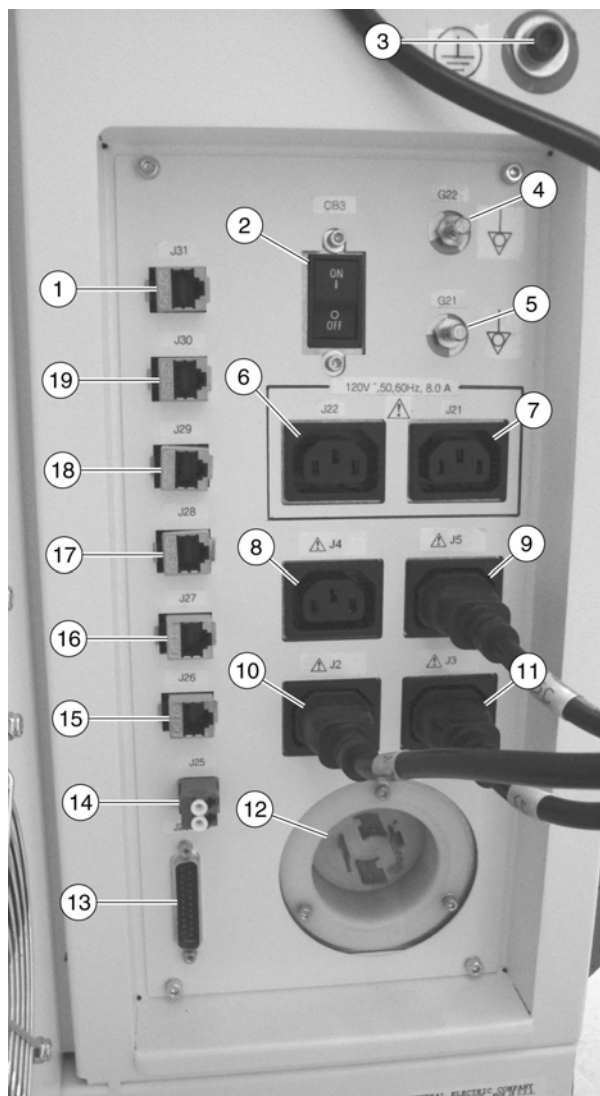
BROADBAND

Broadband is considered the standard network connection for HD.

Broadband connections should use the appropriate ethernet cables for these connections.

The CT system is connected to the network through the Console.

- An ethernet cable (not to exceed 10 feet) should be provided by the customer, and it is used to connect the console to a wall box.
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.



1. J31 RPM
2. CB3 ON/OFF
3. RUN #056 #2 Ground
4. G22
5. G21
6. J22
7. J21
8. J4
9. J5 Media tower
10. J2 Scan Monitor
11. J3 Display Monitor
12. J1 RUN # 053 Console Power
13. J20 RUN #101 Scan INTL (Interlock)
14. J25 RUN # 103 DAS Data (Fiber)
15. J26 HSP (hospital)
16. J27 TGP RUN # 102 LAN (RJ45)
17. J28 UPS
18. J29 AW
19. J30 EKG Cat 5 J45

Figure 8-8 Console Rear Bulkhead

US PROCESS OVERVIEW

The United States network connectivity requirement for this product is broad-band. The US process relies on the Install Specialist to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

CUSTOMER BROADBAND RESPONSIBILITIES

Provide GE Healthcare Project Manager of Installation with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
 - Coordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broadband infrastructure meets GE Healthcare requirements for connection as determined by a mutual assessment with the GE Healthcare Connectivity team.
- IT Contact
 - Complete an equipment assessment with GE Healthcare Connectivity team to determine site readiness for broadband
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - To utilize an Internet Service Provider that supports static routing

Chapter 9

System-Level Safety Tests

You must complete these tests after all options are installed. They cover three safety and leakage current checks:

- Patient Touch Current Test (completed after installation)
- System Ground Resistance Measurement (completed during installation)
- Ground Current Typical (completed after installation - optional)

Section 1.0 Patient Touch Leakage Test

1.1 Time and Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
1	10 mins	20 mins	10 mins

1.2 Tools and Test Equipment

- Standard FE Tool Kit
- Dale 600 Meter (from tool pool; p/n 48-328406G1)
- Date extended length lead (part of p/n 46-328406G1)

1.3 Safety



WARNING



POTENTIAL FOR SHOCK

GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”. FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN ENERGIZED SYSTEM.



NOTICE

Follow ALL required safety and PPE procedures customary for your organization, when working on this product.

1.4 Preparation

- Only trained service personnel should service the GE CT Scanner.
- Footswitch cover must be removed.

1.5 Procedures

- 1.) Move the table to ISO elevation.
- 2.) Remove the footswitch covers and the gantry left side cover.
- 3.) Refer to the Dale 601 Operator's Manual for instructions on the use of the Dale 600 meter for measuring leakage current (or refer to [Figure 9-1](#), for a quick overview)

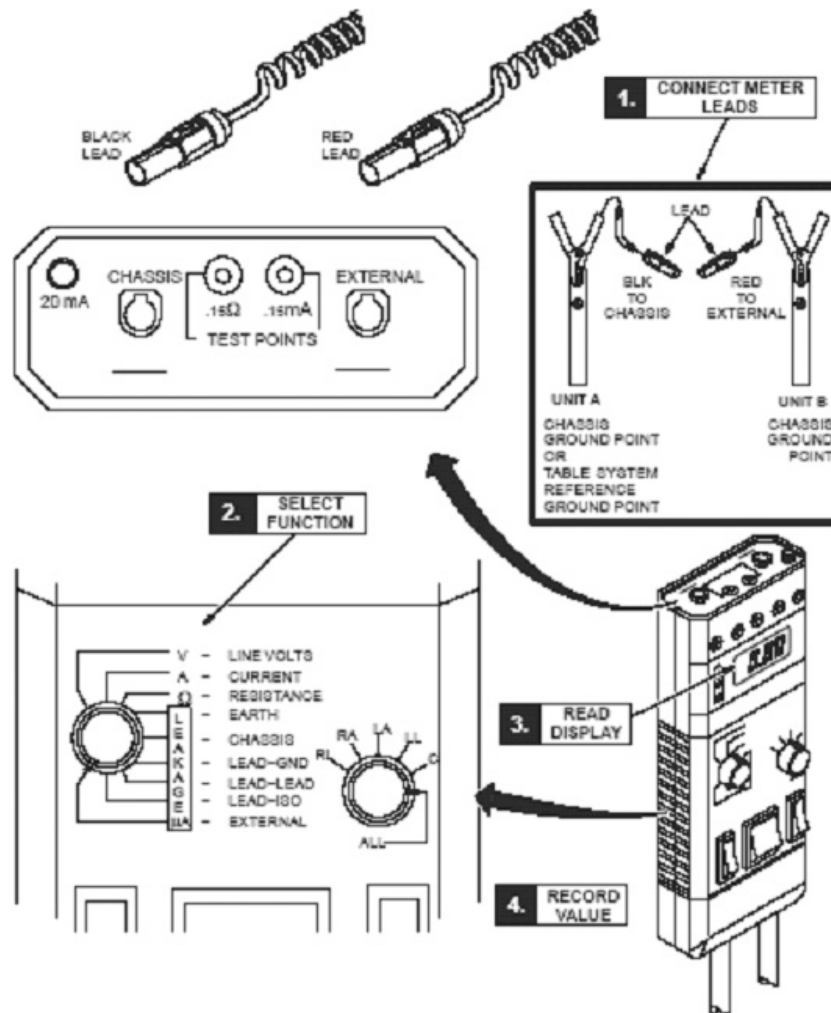


Figure 9-1 Using Dale 600 to Measure Leakage Current

- 4.) Plug the Dale 600 Meter into the outlet on the gantry left side. Confirm the outlet is correctly wired per the three LED indicators on the meter.
- 5.) Connect one end of the shorter, black lead to the chassis plug and connect the other end to the table ground bus.
- 6.) Connect the longer, red test lead (or the longer black lead) to the external plug on the top of the Dale 600 meter.
- 7.) Set the function switch on the Dale meter to *external*. Use the external lead to touch the meter's test terminal, to test that the meter is operational.
- 8.) The black lead is connected to the table base ground bus, and the read lead is connected to the devices (components) under test.

Note: Your meter may have two black leads that are keyed for chassis and the ground connection.

A valid calibration sticker must be present on the meter you use. Record this information on the GE Form e4879.

- 9.) Leakage current is tested with power ON.:

Measurement Area	Current Not to Exceed
Critical care areas (invasive)	10µA CT Systems
General care areas	20µA
Not intended for patient area	50µA

Table 9-1 Maximum Leakage Current Measurements

- 10.) Complete testing between the system reference ground point (table base) to unit reference ground points (i.e. gantry and table, see [Figure 9-3](#)).
- 11.) Test all conductive surfaces and components within patient reach or within 1.9 m (6 ft) of the table and 2.3 m (7.5 ft) above the table.
- Measure at table maximum travels.
 - At some sites, wall outlet cover plates and sinks may become an issue.
- 12.) Test all Optional components such as in-room monitors, injector, overhead monitor suspensions, and table options.

Record these results in the data sheet.

Gnd Bus to	Install <10µA
Any table ground points within the 6' range	
Any gantry ground points within the 6' range	
Injector assembly metal surface	
Boom-in-Room metal surface	
Monitors or metal surface	
Sink or metal surface	
Installed table accessories	
(other)	
(other)	
(other)	

- 13.) **Test conditions:**

- a.) Table max elevation, Min - Max travel
- b.) Patient face up and face down head first, feet first
- c.) Table min elevation 7.2 ft patient access to all conductive surfaces including:
 - * IV poles and tray assembly
 - * Any smart step monitors and stands
 - * Table bearing rails
 - * All other conductive surfaces



NOTICE

Be aware of Static Discharge from: scan window, keypads, displays, touch pads, or other plastic surfaces.

14.) **Procedure Hints:**

- a.) Look for items that have abnormal measurements, high or low.
These could indicate mis-wiring, loose connections, or poor connections due to corrosion, painted surfaces, etc.
- b.) High leakage could indicate a wiring error such as a neutral connected to the ground.
- c.) Fluctuating ground currents could indicate a short, poor connection, or facilities ground problem, causing leakage currents from other areas of the facility to flow through our system grounds.
- d.) Refer to the illustrations below, for measurement visual descriptions.

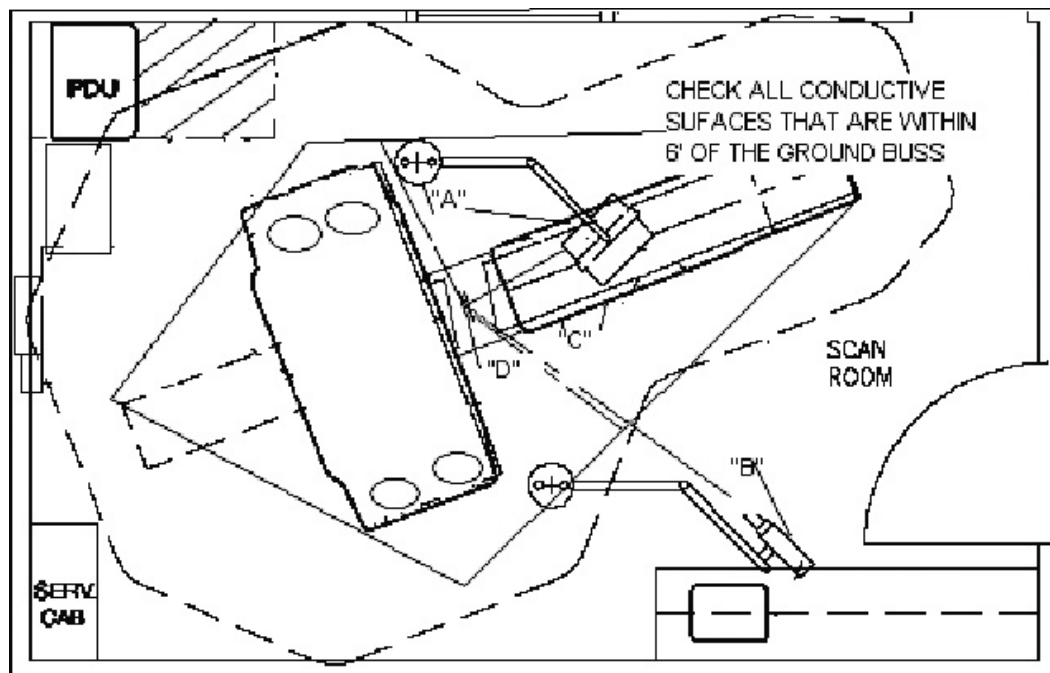


Figure 9-2 CT Leakage Current Map

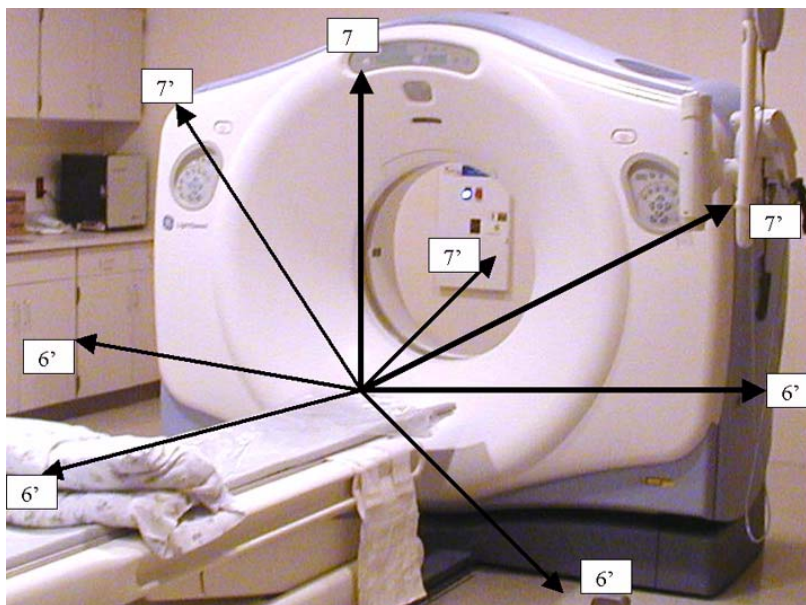


Figure 9-3 Measurement Points

1.6 Ground Resistance Checks Procedure (done during mechanical install)

Refer to the procedures found in Book 1, [Chapter 3](#):

- [Section Section 1.0 - System Continuity \(Mechanical Contractor\)](#)
- [Section Section 2.0 - Site Ground Continuity Check](#)

1.7 Finalization

No finalization required.

Section 2.0 CT System Chassis Leakage Test

2.1 Time & Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
1 Engineer 1 customer electrician		2 hours labor on-site	

Labor time includes:

- Removal of covers
- Leakage test
- Reinstallation of covers
- Recording of site data on the CT System Chassis Leakage Test Completion Form, if required by your state. Forward the completed form to your Project Manager of Installation.

2.2 Tools and Test Equipment

- Standard FE Service Tool Kit
- GE LOTO Kit
- Dale 600 or 601 Leakage Current Meter (from tool pool; p/n 46328406G1)
- Documentation: LOTO PPE in the Equipment Service section of the Service Methods manual.

Note: This procedure was validated only with the Dale 600/601. GE cannot guarantee the accuracy of this procedure if you use another meter.

2.3 Safety



DANGER POTENTIAL FOR ELECTRIC SHOCK
GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”.
THIS TEST IS PERFORMED WITH POWER “ON” - SYSTEM IN STAND-BY!



NOTICE Follow ALL required safety, PPE, and arc-flash procedures customary for your organization, when working on this product.



DANGER POTENTIAL FOR ELECTRIC SHOCK
SERVICING HARDWARE WITH POWER “ON”.

2.4 Procedures

Follow LOTO and other safety procedures found in this manual before starting this procedure.

- 1.) Remove the Gantry top and side covers. Guide rails on each side of the top cover will aid in removal. Refer to Appendix A: [Gantry Cover Removal and Dolly Setup on page 167](#) for more details.

- Note: Fans located on the top cover will make that cover heavier.
- 2.) Power down the console and follow the GEH LOTO procedure.
 - 3.) Remove the Gantry front cover, see cover procedure in Book 1.
 - 4.) Remove the Table foot switch top cover to gain access to the ground cables and ground bar.
- Note: Do NOT disconnect ANY grounds at this time.
- 5.) Confirm that all system grounds are securely attached to the system ground buss and NOT the Table base.
 - 6.) The electrician will remove all external electrical connections made during installation, including:
 - a.) Main system ground at PDU
 - b.) Power feeder flex connection at PDU
 - c.) Room door interlocks and room warning light connections
 - d.) Any and all other external ground connections to the system.
 - 7.) Confirm that all external gantry, table, console, and PDU connections have been removed.
- Note: Some wires such as the room warning light may have external power and wire nuts, which should be installed to protect from arching.
- 8.) Follow the LOTO procedure for re-energizing power and boot to application level. If not already done, remove the footswitch assembly top cover while the system boots.
 - 9.) Plug the Dale 600 / 601 Leakage Current Meter into one of the outlets on the gantry.
 - 10.) Connect the meter leads to the meter as follows:
 - a.) Connect one end of the shorter black lead to the chassis plug and the other to the table ground bus.
 - b.) Connect the longer red test lead to the external plug on top of the Dale 600 / 601 meter.
 - 11.) Set the function switch on the Dale 600 / 601 meter to EXTERNAL. Using the external lead, touch the meter's test terminal to confirm that the meter is operational.
- Note: For more detailed information, refer to the Dale 600 / 601 Operator's Manual or see [Figure 9-1](#), for a quick overview.
- With the system at application-level and all components functional, test the system ground wires as follows:
- a.) Remove a system ground wire.
 - b.) Test that system ground wire.
 - c.) Replace that system ground wire.
 - d.) Repeat, testing all system ground wires ONE-AT-A-TIME. A list of the system ground wires appears in [Table 9-2](#).
- Note: The measured leakage current must not exceed 5 MA in any ground wire.

Components	Results	Date	Tester
PDU Leakage 1/0			
Gantry Leakage 1/0			
Console Leakage #2			
Table Leakage #10			
Option Grounds, if present			

Table 9-2 System Ground Measurements

- 12.) After completing all tests, follow the LOTO procedure to power down the system.
- 13.) The electrician will reinstall all electrical connections, conduits, cables, and wires removed in Step 6 and will secure all connections per NEC code. Check that all connections are securely tightened.
- 14.) Reinstall all removed system covers, except for the gantry right-side cover, located by the service switch panel.

SYSTEM POWER-UP AND TEST

Note: Complete this section of the installation manual on-site.

- 1.) Follow the LOTO procedure for re-energizing power.
- 2.) Turn-on the gantry service switches and power up the console.
- 3.) Check that no cables remain in the gantry rotating path.
- 4.) Return cover dollies to storage areas.
- 5.) Check that the table controls and footswitch function properly.
- 6.) Re-test the system by completing a system functional scan. If installed, be sure to test the room warning light and the door interlock at this time.
- 7.) Complete the CT System Chassis Leakage test, if required by your state, and forward the completed form to your Project Manager of Installation. (This form is located on the Service CD.)

2.5 Finalization

No finalization required.

Appendix D

System Configuration Data Sheets

Section 1.0 Requirements

- Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their CT scanner and their network and filming expectation reduces the time required to reconfigure the system.
- [Table D-1 Manual Film Composer Options, on page 301](#)
 - [Table D-2 System Network Configuration, on page 302](#)
 - [Table D-3 Networking Application \(Image transfer\) Configuration, on page 303](#)
 - [Table D-4 DASM Laser Camera Configuration, on page 303](#)
 - [Table D-5 DICOM Print Camera Configuration, on page 304](#)
 - [Table D-6 DICOM Print Camera Advanced Configuration, on page 304](#)

Section 2.0 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS	
Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table D-1 Manual Film Composer Options

Section 3.0 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* Ask customer or check log	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table D-2 System Network Configuration

HOST ETHERNET ADDRESS

_____ : _____ : _____ : _____ : _____

Section 4.0 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table D-3 Networking Application (Image transfer) Configuration

Section 5.0 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table D-4 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION	
Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table D-5 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCED CONFIGURATION	
Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table D-6 DICOM Print Camera Advanced Configuration

Appendix E

Informational Support Procedures

Section 1.0 Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5032	Alternating Current
	335-1	Three-phase Alternating Current
	335-1	Three-phase Alternating Current with neutral conductor
		Direct Current
	417-5019	Protective Earth (Ground)
	348	Attention, consult ACCOMPANYING DOCUMENTS
	417-5008	OFF (Power: disconnection from the mains)
	417-5007	ON (Power: connection to the mains)
		Warning, HIGH VOLTAGE
		Emergency Stop

Table E-1 Symbols

SYMBOL	PUBLICATION	DESCRIPTION
		Type B
	417-5339	X-ray Source Assembly Emitting
	417-5009	Standby
		Start
		Table Set
		Abort
		Intercom
		(on Operator Console) Power On: light on Standby: light off

Table E-1 Symbols

Section 2.0 ESD Grounding Points

The gantry has defined ESD attachment points that allow easy attachment for banana clip and alligator clip style connections. There are points on the stationary frame on each side of the gantry and rotating assembly behind the DAS plate. The stationary frame locations should always be used if the length of your ESD strap is sufficient.

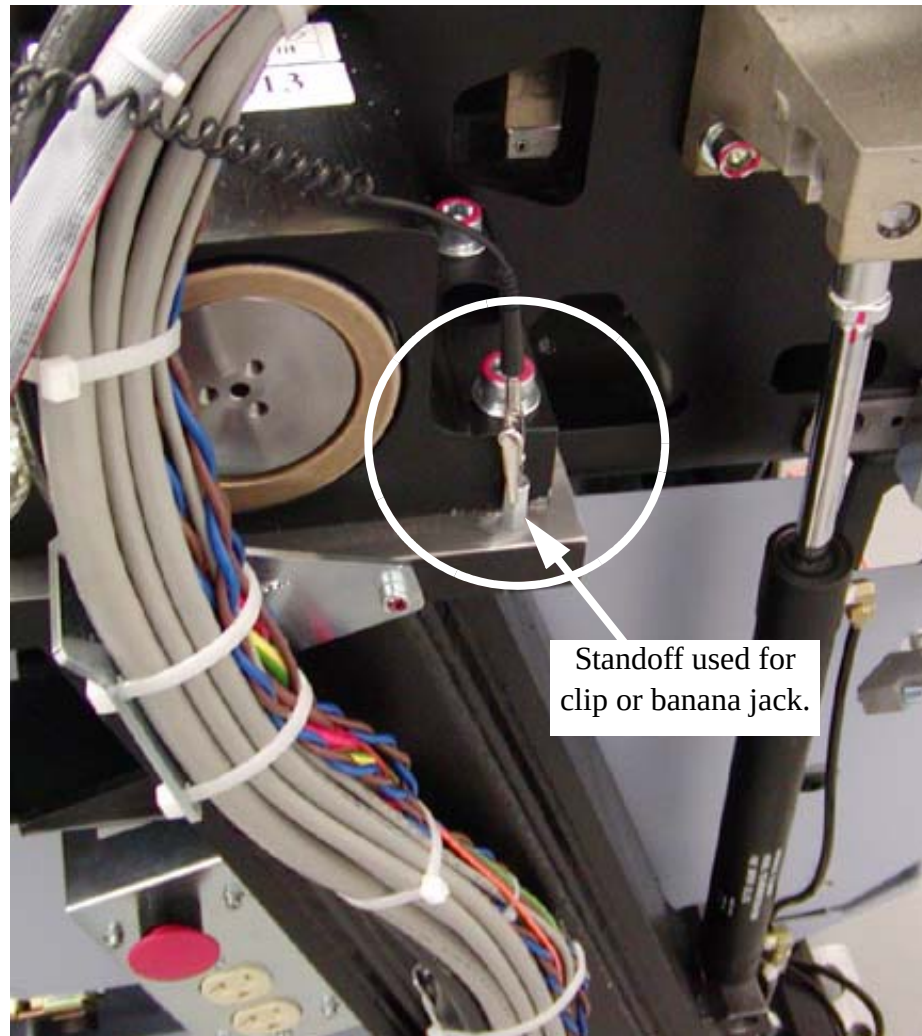


Figure E-1 Gantry Right Side Clip Example

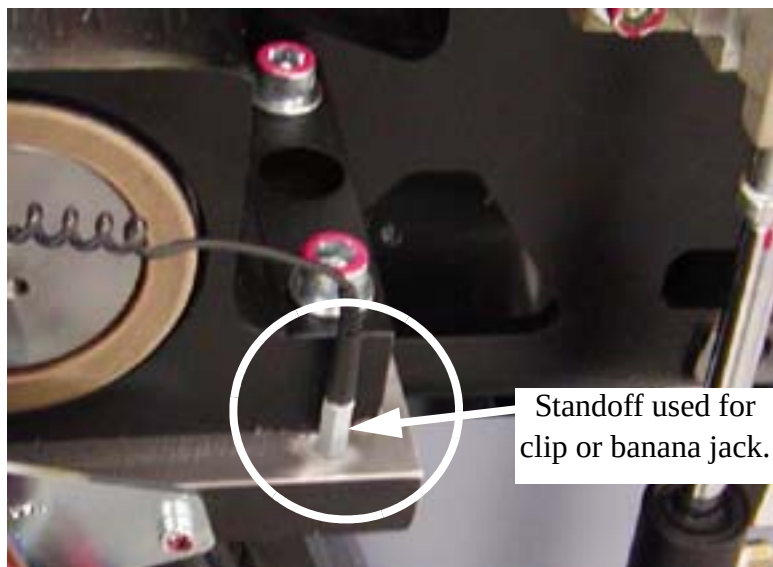


Figure E-2 Gantry Right Side Banana Jack Example

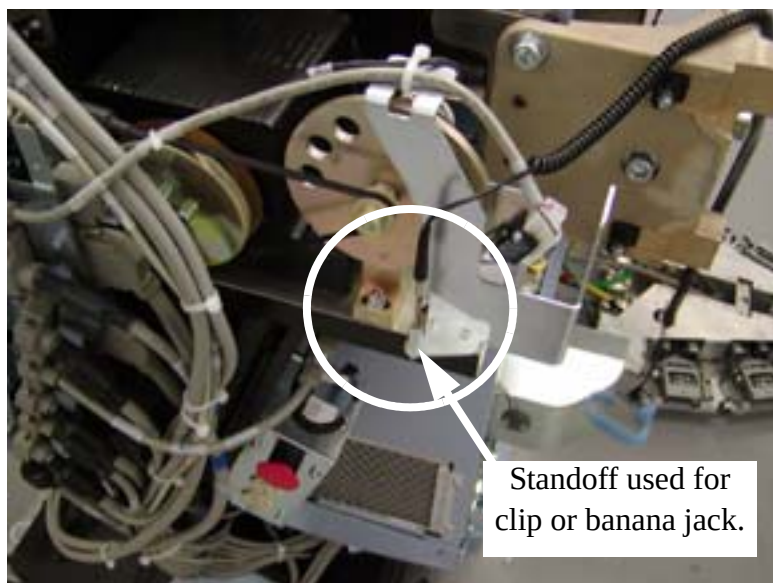


Figure E-3 Gantry Left Side Connection



Figure E-4 Rear of DAS Plate ESD Connection Points

Section 3.0 Manually Mount/Unmount a CD

3.1 Introduction

This appendix contains the following procedures that may be useful during a software load:

- CDROM — Mount Process
- CDROM — Unmount Process

3.2 Procedures

2.5.1 Preparation

- 1.) Insert a CD into the DVD ROM drive.
- 2.) Open a Shell window.
- 3.) Type: **su -** ENTER
- 4.) Type the password and press ENTER

2.5.2 CDROM — Manual Mount CD Process

If login was successful, continue with manually mounting the CD:

- 1.) Type: **cd /** ENTER
- 2.) Type: **mount /mnt/cdrom** ENTER
- 3.) Type: **cd /mnt/cdrom** ENTER
- 4.) The CD has now been mounted. To see what is on it, type:
ls /mnt/cdrom ENTER
- 5.) Type: **./copy_rpms** ENTER

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

2.5.3 CDROM — Manual Unmount CD Process

Unmount the CD in order to remove it from the drive; do this by typing the following:

- 1.) Type: **cd /** ENTER
- 2.) Type: **umount /mnt/cdrom** ENTER
- 3.) Type: **eject** ENTER
- 4.) Remove the CD from the DVD-ROM drive and close the drive.

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